

Beyond Realized Volatility: Why the IVOL Puzzle May Be a Mirage

Reza Kahibavil

Abstract

This paper revisits the idiosyncratic volatility (IVOL) puzzle the counterintuitive finding that stocks with higher idiosyncratic risk earn lower future returns by evaluating whether improved forecasting and decomposition of volatility can reconcile this anomaly with rational asset pricing. Using U.S. equity data from 1962 to 2023, I construct six forecasts of monthly IVOL: a baseline ARMA(1,1) model, three earnings-announcement-adjusted variants, a max-adjusted forecast, and a cross-sectional model based on firm characteristics. While the IVOL puzzle persists in value-weighted portfolios, it weakens or disappears in equal-weighted portfolios and in cross-sectional Fama-MacBeth regressions using forecasted IVOL. Decomposing volatility into event-driven (EAD) and non-event-driven (NOEAD) components reveals that only NOEAD volatility carries a significantly negative return premium, suggesting that investors do not penalize informational volatility shocks. I further introduce a novel diagnostic ratio the squared residual from a stock's maximum daily return scaled by its max-adjusted monthly volatility which isolates episodic volatility spikes. Portfolios sorted on this ratio show that the IVOL puzzle concentrates in stocks with transient, noisy volatility rather than persistent risk. Overall, the findings suggest that the IVOL puzzle stems from volatility mismeasurement rather than irrational pricing and highlight the importance of filtering out non-informational volatility to better understand return–risk relations.

Keywords: *Idiosyncratic volatility, asset pricing anomalies, volatility forecasting, earnings announcements, lottery stocks, measurement error*

1. Introduction

The relationship between risk and return lies at the heart of asset pricing theory. Under classical models such as the CAPM and its multifactor extensions, investors are expected to be rewarded for bearing systematic risk, while idiosyncratic risk, being diversifiable, is assumed to be unrewarded. However, a puzzling empirical regularity has persisted in recent decades: stocks with higher idiosyncratic volatility tend to earn **lower**, not higher, future returns. This counterintuitive result, known as the **idiosyncratic volatility (IVOL) puzzle**, poses a challenge to foundational asset pricing models and has sparked extensive debate across both rational and behavioral schools of thought.

The IVOL puzzle is more than a theoretical curiosity, it has practical consequences for portfolio construction, risk management, and the development of multifactor models. As factor-based investing strategies become increasingly popular in institutional settings, understanding how and why certain anomalies persist becomes critical. If idiosyncratic risk truly earns negative premia, traditional optimization and risk assessment frameworks may systematically misallocate capital. For regulators and academics alike, resolving the IVOL puzzle touches on core questions of market efficiency, the nature of risk compensation, and the potential role of investor behavior in price formation.

A seminal contribution by Ang, Hodrick, Xing, and Zhang (2006) documents the negative cross-sectional relation between past IVOL and future stock returns, even after controlling for known risk factors. Their follow-up study (Ang et al., 2009) confirms the puzzle across 23 international markets, suggesting a global phenomenon. Subsequent research has attempted to explain the puzzle through both rational and behavioral lenses. On the rational side, Fu (2009) argues that the expected idiosyncratic volatility, not the realized one, should be priced, and shows that conditional IVOL (estimated via EGARCH) is positively related to future returns. However, Han and Lesmond (2011) point out that illiquidity may confound IVOL estimates, while Hou and Loh (2016) show that much of the anomaly is concentrated in stocks with high trading frictions.

On the behavioral side, Bali, Cakici, and Whitelaw (2011) provide evidence that the IVOL puzzle is driven by stocks with lottery-like features, i.e., those that have high skewness and extreme daily returns. The preference for lottery-like payoffs, also discussed in Kumar (2009) and Bali, Brown,

and Murray (2014), suggests that investors may overpay for stocks with the potential for extreme upside, despite their long-run underperformance. Stambaugh, Yu, and Yuan (2015) introduce the concept of mispricing-related characteristics and show that anomalies like IVOL are stronger among stocks with high limits to arbitrage, suggesting that behavioral biases persist due to frictions that prevent rational investors from correcting mispricing.

While these studies offer a wide range of explanations, consensus on the source of the IVOL puzzle remains elusive. More recently, Bekaert, Bergbrant, and Kassa (2025) introduce a forecasting-based perspective: they argue that poor volatility forecasts, especially from simple martingale models, lead to persistent overestimation of future IVOL. Their results show that when forecast accuracy improves, using ARMA-based models, the strength of the puzzle diminishes substantially. Their insight reframes the anomaly not as a failure of pricing per se, but as a reflection of systematic forecast error.

Building on this foundation, my study extends the literature in three major directions. First, I construct six alternative methods to forecast one-month-ahead IVOL. These include a baseline ARMA(1,1) model, three earnings announcement-adjusted forecasts that exclude return windows of ± 2 , ± 3 , and ± 5 days, a MAX-adjusted model that removes the month's maximum daily return, and a cross-sectional Fama-MacBeth model that uses firm characteristics such as MAX, momentum, illiquidity (Amihud ratio), and size. These models capture distinct features of volatility, informational vs. noisy, episodic vs. persistent, and allow for a systematic evaluation of whether the puzzle arises from how volatility is measured and predicted.

Second, I investigate the role of forecast error by decomposing prediction errors into **overestimated** and **underestimated** groups. In line with Bekaert et al. (2025), I find that the IVOL puzzle is concentrated among overestimated forecasts: stocks for which predicted IVOL exceeds realized volatility earn significantly lower returns. In contrast, when IVOL is underestimated, the return relationship reverses, high-IVOL stocks actually outperform their low-IVOL counterparts. This asymmetry underscores a key insight: the IVOL puzzle is not simply a pricing failure but is tightly linked to the nature and direction of forecasting bias.

Third, I isolate the source of volatility by distinguishing between idiosyncratic risk related to earnings announcements and that arising from non-event periods. I define EAD-IVOL as volatility concentrated in windows around earnings releases, and NOEAD-IVOL as the residual

idiosyncratic volatility occurring outside these windows. Fama-MacBeth regressions show that NOEAD-IVOL is consistently and significantly negatively related to future returns, while EAD-IVOL is uncorrelated or even weakly positively related to returns. These findings suggest that the market penalizes volatility with less informational content and appropriately ignores volatility that reflects public information events.

To capture the structure of volatility within each month, I introduce a new diagnostic variable, the **MAX IVOL Ratio**, which measures the share of monthly IVOL explained by a single extreme daily residual. This ratio proxies for whether a stock's volatility is driven by a single rare event or is more evenly distributed across days. I find that stocks with higher MAX IVOL Ratios (more concentrated volatility) tend to earn **higher future returns**, especially in value-weighted portfolios. In contrast, the IVOL puzzle remains pronounced among stocks with more diffuse volatility. Even after controlling for volatility structure, however, I find that the overall level of IVOL remains negatively associated with returns, suggesting that markets may reward exposure to rare-event risk but penalize persistent noise.

Together, these results offer a comprehensive reinterpretation of the IVOL puzzle. I show that the anomaly is **conditional**, not universal, it arises from overestimation, non-event-driven volatility, and persistent residual noise. My approach unifies and extends prior work by integrating volatility forecasting, event filtering, diagnostic signals, and behavioral conditioning into a coherent framework. Unlike earlier studies that rely on single-model volatility estimates or narrow conditioning variables, my methodology reveals the underlying components that make IVOL appear to be mispriced.

This paper contributes to the asset pricing literature by deepening our understanding of how investors process different forms of idiosyncratic risk. It highlights that not all IVOL is created equal, its informational source, forecastability, and structure matter for pricing. It also refines the use of forecasting in anomaly studies, suggesting that forecast-based diagnostics can uncover when apparent anomalies are actually artifacts of model specification. Beyond theory, these findings offer practical insights for portfolio managers, who may avoid costly misallocations by distinguishing between risk-driven and noise-driven volatility signals. The framework introduced here can also be applied to other cross-sectional anomalies, such as momentum or mispricing-

based factors, to better understand the dynamics of risk perception and investor behavior in financial markets.

2. Literature Review and Conceptual Framework

2.1 Overview of the IVOL Puzzle

The idiosyncratic volatility (IVOL) puzzle is a widely studied yet unresolved anomaly in empirical asset pricing. According to foundational asset pricing theories, such as the Capital Asset Pricing Model (CAPM) by Sharpe (1964) and Lintner (1965), only systematic risk should be priced, while firm-specific (idiosyncratic) risk can be diversified away in large portfolios. Consequently, idiosyncratic volatility should not be systematically related to expected returns.

Contrary to this prediction, Ang, Hodrick, Xing, and Zhang (2006) provide compelling evidence that stocks with higher idiosyncratic volatility, measured as the standard deviation of residuals from a Fama-French three-factor model, earn significantly **lower** subsequent returns. Their study shows that stocks in the highest IVOL quintile underperform those in the lowest quintile by approximately 1% per month, a magnitude difficult to reconcile with standard risk-based models. This negative return premium persists even after adjusting for known factors such as size, book-to-market, and momentum.

Subsequent work reinforces the robustness and global scope of the puzzle. Ang et al. (2009) replicate their findings across 23 developed and emerging markets, and similar results are documented by Hou and Loh (2016), Jiang, Xu, and Yao (2009), and Wei and Zhang (2005). These studies confirm that the negative IVOL–return relationship is not sample-specific or limited to U.S. equities, but appears to be a persistent feature of asset returns across a wide range of contexts.

2.2 Risk-Based Explanations

One line of research seeks to rationalize the IVOL puzzle within traditional risk-based frameworks by arguing that high-IVOL stocks may be exposed to risks not captured by standard models. Campbell et al. (2001) associate high idiosyncratic volatility with firms in financial distress or undergoing significant operational changes, both of which may involve unpriced or nonlinear risks. Fu (2009) argues that it is **expected** idiosyncratic volatility, not realized IVOL, that matters.

Using a conditional EGARCH model, he finds that conditional IVOL is positively related to returns, suggesting that earlier findings may have conflated transitory volatility with priced risk.

Other studies attribute the puzzle to **tail risk** or downside exposure. Huang, Liu, Rhee, and Zhang (2010) show that high-IVOL firms often exhibit greater crash risk, while Conrad, Dittmar, and Ghysels (2013) highlight the failure of traditional factor models to capture dynamic volatility exposures. More broadly, Lettau, Maggiori, and Weber (2014) argue that standard linear factor models may overlook higher-moment risks that are priced in equilibrium.

A different subset of risk-based explanations focuses on **market frictions**. Han and Lesmond (2011) find that illiquidity and sparse trading inflate volatility estimates, particularly for microcap and thinly traded firms. These stocks often exhibit both high IVOL and low returns, which can create a spurious correlation. Similarly, Hou and Moskowitz (2005) show that firms with low analyst coverage and slow information diffusion exhibit return patterns that may confound volatility-based tests. While these explanations may partially account for the puzzle, the IVOL anomaly typically survives even after controlling for liquidity measures, turnover, and institutional ownership (Bali and Cakici, 2008).

2.3 Behavioral Explanations

An influential alternative perspective attributes the IVOL puzzle to **investor behavior** rather than omitted risk factors. Several studies argue that retail investors exhibit a preference for stocks with lottery-like characteristics, low price, high volatility, and positively skewed return distributions. Kumar (2009) provides direct survey and trading evidence that such stocks are disproportionately favored by individual investors with lower financial literacy and higher gambling preferences.

Bali, Cakici, and Whitelaw (2011) develop the MAX variable, defined as the maximum daily return in the past month, to proxy for lottery-like behavior. They find that the IVOL puzzle is concentrated in stocks with high MAX values, suggesting that extreme one-day returns attract speculative attention, leading to overvaluation and poor future performance. Similarly, Boyer, Mitton, and Vorkink (2010) and Mitton and Vorkink (2007) model preferences for skewness in investor utility functions and show that such preferences can produce pricing anomalies similar to the IVOL puzzle.

Another stream of behavioral literature focuses on **mispricing and limits to arbitrage**. Stambaugh, Yu, and Yuan (2015) find that the IVOL puzzle is most pronounced in stocks with high mispricing scores and characteristics indicating limits to arbitrage, such as high idiosyncratic volatility, low institutional ownership, and high transaction costs. Green and Hwang (2012) argue that speculative stocks draw more investor attention and disagreement, making their prices more sensitive to sentiment and harder to arbitrage. Supporting this view, Zhang (2006) and Fong, Gallagher, and Lee (2014) find that the IVOL puzzle is more severe during high-sentiment periods, when overpricing is more likely to go unchecked.

Together, these behavioral explanations suggest that the IVOL puzzle arises from a combination of investor overreaction to rare payoffs, preference for skewness, and frictions that prevent arbitrageurs from correcting mispricing. These mechanisms provide a coherent narrative for the persistence and cross-sectional variation of the puzzle, especially among small-cap, low-price, and high-turnover stocks.

2.4 Forecasting-Based Approaches to the IVOL Puzzle

The recent literature has increasingly turned its attention to the role of **volatility forecasting** in explaining the IVOL puzzle. A key advance in this direction is made by Bekaert, Bergbrant, and Kassa (2025), who argue that much of the negative relationship between idiosyncratic volatility and expected returns arises from **poor volatility forecasts**, rather than from pricing inefficiencies per se. Using nearly 80 million daily returns from over 19,000 U.S. firms between 1960 and 2023, they evaluate the out-of-sample performance of several time-series models for predicting next-month idiosyncratic variance. Their results show that the commonly used **martingale model** performs particularly poorly, especially for stocks with extreme return patterns and poor liquidity. In contrast, a simple **ARMA(1,1)** model provides the lowest root-mean-squared prediction error (RMSE) for nearly half the firms.

More importantly, Bekaert et al. demonstrate that once ARMA(1,1) forecasts are used in return regressions, the IVOL puzzle **largely disappears**. They show that the puzzle is primarily driven by a small subset of stocks for which the martingale model significantly **overpredicts** future realized volatility. These cases are heavily associated with firms that have wide bid-ask spreads, high maximum daily returns, and low trading volume, stocks that also figure prominently in the behavioral finance literature as candidates for speculative overpricing (Kumar, 2009; Green and

Hwang, 2012). The implication is that **forecasting error**, especially overestimation, may be a key driver of the anomaly.

This perspective builds on earlier work such as Fu (2009), who argues that **expected IVOL**, rather than realized IVOL, should be priced. Using conditional volatility estimates from EGARCH models, Fu finds that expected IVOL is positively related to returns, implying that mismeasurement of risk may distort empirical inference. However, Fu's analysis stops short of comparing alternative forecast models or linking forecast accuracy to the pricing results. More recent studies, such as Zhang, Zhang, and Zhang (2020), introduce machine learning techniques to improve volatility forecasts and document meaningful variation in the IVOL–return relationship depending on model choice.

Han, Huang, and Wang (2021) extend this forecasting agenda by demonstrating that macroeconomic conditions affect IVOL predictability. Their deep learning approach shows that during economic expansions, IVOL is more persistent and forecastable, whereas in recessions, its dynamics become noisier. These results suggest that **state-dependent variation in forecast quality** may also shape the pricing of idiosyncratic risk. Taken together, these studies underscore the importance of treating IVOL not as an ex-post realized variable, but as a **forecasted quantity** whose accuracy can materially affect asset pricing conclusions.

2.5 Gaps in the Literature

Despite recent progress, several important gaps remain in the literature on IVOL and its pricing implications.

First, current forecast-based approaches typically rely on raw return data and assume a homogeneous volatility process across time. They do not account for **return-generating events**, such as earnings announcements, which have been shown to produce predictable spikes in volatility (Beaver, 1968; DellaVigna and Pollet, 2009). If earnings-related volatility is expected and tied to the firm's information environment, it may not carry the same risk premium, or mispricing, as residual, non-event-driven volatility. Yet no major study has systematically **decomposed IVOL into earnings-related and non-earnings-related components** to test this hypothesis.

Second, there has been little effort to evaluate how **forecast error directionality**, whether forecasts over- or underpredict realized IVOL, affects return patterns. Bekaert et al. (2025) hint at this issue by showing that extreme overestimates of IVOL drive the puzzle, but they do not explicitly separate the return implications of over- versus underestimation across the cross-section. This is surprising given that behavioral theory suggests that **investor overreaction** to anticipated risk (e.g., overestimating uncertainty) could drive prices below fundamentals, while **underestimation** may reflect informational frictions or delayed learning rather than mispricing (Hong and Stein, 1999).

Third, while numerous studies document a correlation between IVOL and lottery-like features, such as high MAX returns (Bali, Cakici, and Whitelaw, 2011), skewness (Boyer, Mitton, and Vorkink, 2010), or price levels (Kumar, 2009), none have introduced a parsimonious metric that captures the **structure** of volatility. Specifically, the distinction between **volatility that is concentrated in one extreme day** versus **volatility that is diffuse across the month** remains largely unexplored. This is a critical gap: volatility concentrated in rare, high-impact events (such as earnings surprises or speculative spikes) may carry different pricing implications than volatility arising from ongoing noise or trading frictions.

Fourth, many empirical studies continue to rely on **realized IVOL** without engaging with the underlying dynamics that drive it. While machine learning and time-series models have improved forecasting accuracy (Gu, Kelly, and Xiu, 2020), the majority of research still treats IVOL as an outcome rather than a **forecasting target** that can be decomposed, conditioned, or explained. As a result, we lack a framework that combines **model-based forecasts**, **event-time conditioning**, and **volatility structure diagnostics** to better understand the IVOL puzzle.

Finally, there is limited attention to the **economic interpretation** of volatility structure. While measures such as kurtosis or realized semivariance have been proposed (Barndorff-Nielsen et al., 2010), they are often statistically noisy or difficult to interpret economically. A diagnostic that links volatility concentration to firm characteristics and return outcomes would allow researchers to distinguish between **informed volatility** (e.g., from earnings shocks) and **uninformed noise** (e.g., from transient sentiment), which in turn could reveal whether the market systematically penalizes one type more than the other.

These gaps highlight the need for a comprehensive empirical strategy that moves beyond traditional realized IVOL measures and embraces **forecasting accuracy**, **event-driven decomposition**, and **volatility structure diagnostics** as central elements in evaluating whether and when the IVOL puzzle truly reflects mispricing.

3. Conceptual Framework and Research Design

3.1 Forecasting IVOL: Six Empirical Models

A central premise of this paper is that the observed relationship between idiosyncratic volatility (IVOL) and future returns depends critically on how IVOL is measured and, more importantly, **forecasted**. If forecast error, especially overestimation of future IVOL, drives the apparent anomaly, then it is essential to compare and contrast multiple forecasting frameworks. I design and estimate six distinct models to forecast one-month-ahead IVOL, each targeting a specific theoretical mechanism: persistence of volatility, informational versus non-informational shocks, and episodic volatility patterns.

All forecasts are generated using **daily data**, and each model estimates volatility over a **rolling window of 120 months** (10 years), with a **minimum requirement of 36 months** to ensure stability. The forecasts are for **next month's IVOL**, defined either as the standard deviation or the root mean square of daily residuals from a Fama-French 3-factor model.

To construct firm-level idiosyncratic volatility measures, I begin by estimating daily residuals from the Fama-French three-factor model using rolling regressions. For each firm-month, I regress daily excess returns on the contemporaneous daily Fama-French market, size (SMB), and value (HML) factors. Specifically, for each stock i on day d , the model is estimated as:

$$r_{i,d,t} - r_{f,d,t} = \beta_{i,MKT}MKT_{m,d,t} + \beta_{i,SMB}SMB_{d,t} + \beta_{i,HML}HML_{d,t} + \varepsilon_{i,d,t} \quad (1)$$

where $r_{i,d,t}$ is the return of stock i on day d in month t , $r_{f,d,t}$ is the risk-free rate, $MKT_{m,d,t}$ is the market return, and $\varepsilon_{i,d,t}$ is the estimated idiosyncratic residual. The regression is run separately for each stock within each calendar month, provided there are at least 14 valid daily observations. The resulting residuals $\varepsilon_{i,d,t}$ form the basis for the monthly idiosyncratic volatility calculations described below.

Before implementing the forecasting models, I compute realized idiosyncratic volatility (IVOL) for each firm using two complementary methods based on daily residuals from the Fama-French three-factor model. In the first approach, IVOL is defined as the **standard deviation** of daily residuals within a calendar month, following the standard practice in the literature. In the second approach, used in Bekaert et al. (2025) and adopted in my forecasting framework, I define IVOL as the **annualized square root of the sum of squared residuals**, emphasizing the cumulative impact of daily idiosyncratic shocks.:

$$IVOL_{i,t}^{std} = \sqrt{\sum_{d=1}^{n_{i,t}} \frac{\varepsilon_{i,t}^2}{n_{i,t} - 1}} \quad (2)$$

$$IVOL_{i,t}^{rss} = \sqrt{\frac{252}{n_{i,t}} \sum_{d=1}^{n_{i,t}} \varepsilon_{i,t}^2} \quad (3)$$

where $\varepsilon_{i,d}$ is the residual return of stock i on day d , and $n_{i,t}$ is the number of valid trading days in month t . The standard deviation-based measure reflects average daily residual volatility within the month, while the annualized RSS-based measure captures total volatility scaled to a 252-day trading year. Both measures are used throughout the forecasting models to assess robustness across alternative volatility definitions.

Model 1: ARMA(1,1) on Realized IVOL (Baseline Forecast)

The baseline forecast model uses an ARMA(1,1) time-series process to capture the persistence in realized idiosyncratic volatility. Bekaert et al. (2025) demonstrate that this model outperforms both martingale and simple historical average models in terms of root-mean-squared prediction error.

Let $IVOL_{i,t}$ denote the realized IVOL for stock i in month t . The model is estimated as:

$$IVOL_{i,t} = \alpha_i + \phi_i IVOL_{i,t-1} + \theta_i \varepsilon_{i,t-1} + \varepsilon_{i,t} \quad (4)$$

The one-step-ahead forecast is:

$$\widehat{IVOL}_{i,t+1}^{ARMA} = \alpha_i + \phi_i IVOL_{i,t} + \theta_i \hat{\varepsilon}_{i,t} \quad (5)$$

This model captures short-term persistence and autoregressive mean reversion while allowing for moving-average corrections of prior shocks.

Model 2–4: EA-Adjusted ARMA Models ([−2,+2], [−3,+3], [−5,+5])

The next three models account for the informational content of earnings announcements (EAs). Prior studies (Beaver, 1968; Ball and Shivakumar, 2008) show that returns exhibit predictable volatility spikes around EAs. If these are expected by investors, they may not represent true uncertainty and should be filtered out when forecasting IVOL.

To isolate volatility outside of EA windows, I exclude daily returns within $\pm k$ days of an earnings announcement for $k \in \{2,3,5\}$. The daily residuals are estimated from the Fama-French 3-factor model **excluding these windows**, and monthly realized IVOL is redefined accordingly.

The ARMA(1,1) model is then applied to these **EA-adjusted IVOL series**:

$$IVOL_{i,t}^{EA(\pm k)} = \sqrt{\sum_{d \notin EA_k}^{n_{i,t}} \frac{\varepsilon_{i,t}^2}{n_{i,t}^{non_EA} - 1}} \quad (6)$$

for each k-window variant. The forecast becomes:

$$\widehat{IVOL}_{i,t+1}^{EA(\pm k)} = \alpha_i + \phi_i IVOL_{i,t}^{EA(\pm k)} + \theta_i \hat{\varepsilon}_{i,t} \quad (7)$$

These models aim to remove volatility from **informational events**, isolating residual (possibly mispriced) volatility.

Model 5: MAX-Adjusted ARMA Model

The fifth model tests whether the IVOL puzzle is driven by **extreme return shocks**, such as single-day spikes unrelated to fundamentals. Bali et al. (2011) introduce the MAX variable to capture such behavior. Here, I extend that insight by filtering out the **maximum daily return each month** and recalculating IVOL based on the remaining days.

Let $d_{i,t}^*$ denote the day with the maximum absolute return for stock i in month t . The MAX-adjusted realized IVOL is:

$$IVOL_{i,t}^{MAX_Adj} = \sqrt{\sum_{d \in d_{i,t}^*} \frac{\varepsilon_{i,t}^2}{n_{i,t} - 2}} \quad (8)$$

The ARMA(1,1) model is then estimated and forecasted using this adjusted series. This approach tests whether **episodic spikes**, rather than sustained volatility, drive the pricing anomaly.

$$\widehat{IVOL}_{i,t+1}^{MAX_Adj} = \alpha_i + \phi_i IVOL_{i,t}^{MAX_Adj} + \theta_i \hat{\varepsilon}_{i,t} \quad (9)$$

Model 6: Cross-Sectional Forecast via Fama-MacBeth Regressions

Unlike the previous five models, which rely on time-series forecasts, the final model employs a **cross-sectional predictive approach**. Inspired by the literature on expected volatility and firm characteristics (Fu, 2009; Bali et al., 2011; Gu, Kelly, and Xiu, 2020), I construct monthly Fama-MacBeth regressions to predict IVOL using firm-level variables.

For each month t , I run the following cross-sectional regression:

$$IVOL_{i,t+1} = \alpha_i + \beta_1 IVOL_{i,t} + \beta_2 MAX_{i,t} + \beta_3 SIZE_{i,t} + \beta_4 MOM_{i,t} + \beta_5 ILLIQ_{i,t} + \varepsilon_{i,t} \quad (10)$$

The coefficients are then **averaged across a 120-month rolling window** to obtain stable estimates:

$$\bar{\beta}_j = \frac{1}{120} \sum_{s=t-119}^t \beta_j^{(s)} \text{ for } j = 1, 2, 3, 4, 5 \quad (11)$$

The predicted IVOL for month $t + 1$ is:

$$\widehat{IVOL}_{i,t+1}^{CS} = \bar{\alpha}_i + \bar{\beta}_1 IVOL_{i,t} + \bar{\beta}_2 MAX_{i,t} + \bar{\beta}_3 SIZE_{i,t} + \bar{\beta}_4 MOM_{i,t} + \bar{\beta}_5 ILLIQ_{i,t} \quad (12)$$

This model captures **observable risk characteristics** and behavioral preferences that may systematically affect future volatility.

3.2 Decomposing Realized Volatility: EAD vs. NOEAD

An important aspect of volatility often overlooked in empirical asset pricing is the **source** of the return variation. A substantial portion of firm-level volatility arises around information-rich events, particularly quarterly earnings announcements. Prior research (e.g., Beaver, 1968; Ball and Shivakumar, 2008; Dellavigna and Pollet, 2009) has documented that these events are associated

with sharp, temporary increases in trading activity and return volatility, reflecting the market's response to new fundamental information. However, these spikes are largely anticipated and may not represent uncertainty in the way that traditional risk-based theories define it. If investors expect high volatility around such events, they may not penalize it with a return discount. As such, pooling event-related and non-event-related volatility into a single measure of IVOL may obscure meaningful cross-sectional pricing dynamics.

To disentangle these two sources, I decompose monthly realized idiosyncratic volatility into **EAD IVOL**, driven by returns within earnings announcement windows, and **NOEAD IVOL**, which captures the remaining volatility outside of those periods. This decomposition allows me to test whether the IVOL puzzle is predominantly driven by volatility surrounding anticipated information events or by unexplained fluctuations during non-event periods.

The decomposition is based on daily residuals from the Fama-French three-factor model. Let $\varepsilon_{i,d}$ denote the residual return of firm i on trading day d , and let month t include $n_{i,t}$ total valid trading days. Suppose $D_{i,t}^{EAD} \subseteq \{1, \dots, n_{i,t}\}$ is the set of dates within a symmetric window surrounding an earnings announcement, such as ± 2 , ± 3 , or ± 5 days. The number of days in the EAD window is $|D_{i,t}^{EAD}|$, and the number of days outside the window is $n_{i,t} - |D_{i,t}^{EAD}|$.

The **EAD IVOL** is then defined as:

$$EAD_IVOL_{i,t}^{std(w)} = \sqrt{\sum_{d \in D_{i,t}^{EAD}} \frac{\varepsilon_{i,d}^2}{|D_{i,t}^{EAD}| - 1}} \quad (13)$$

$$EAD_IVOL_{i,t}^{rss(w)} = \sqrt{\frac{252}{|D_{i,t}^{EAD}|} \sum_{d \in D_{i,t}^{EAD}} \varepsilon_{i,d}^2} \quad (14)$$

where w indicates the width of the event window (i.e., 5, 7, or 11 trading days). If a firm does not have an earnings announcement during month t , then $EAD_IVOL_{i,t}^{(w)}$ is set to zero. The **NOEAD IVOL** is computed using the residuals from non-event days:

$$EAD_IVOL_{i,t}^{std(w)} = \sqrt{\sum_{d \notin D_{i,t}^{EAD}} \frac{\varepsilon_{i,d}^2}{n_{i,t} - |D_{i,t}^{EAD}| - 1}} \quad (15)$$

$$EAD_IVOL_{i,t}^{rss(w)} = \sqrt{\frac{252}{n_{i,t} - |D_{i,t}^{EAD}|} \sum_{d \notin D_{i,t}^{EAD}} \varepsilon_{i,d}^2} \quad (16)$$

This decomposition ensures that the total realized idiosyncratic variance is the sum of squared EAD and NOEAD residuals, scaled appropriately for annualization. These separate series allow me to test whether only the **non-event component of IVOL is penalized** in returns, an implication consistent with the notion that markets treat unexpected or noisy volatility differently than that arising from fundamental information flows.

3.3 Diagnosing Volatility Structure: The MAX IVOL Ratio

Beyond the source of volatility, its **internal structure** may also affect how it is priced. Specifically, monthly volatility may either be driven by consistent daily fluctuations or be concentrated in a single extreme return, such as a sharp price jump or drop on one day. Traditional volatility measures like standard deviation or root mean squared residuals fail to differentiate between these two cases. Yet, they may be **economically distinct**. A spike caused by a one-time noise event (e.g., a liquidity shock, a rumor, or a brief pricing error) may not indicate sustained uncertainty. If such episodes disproportionately inflate IVOL and contribute to the puzzle, then a volatility **structure diagnostic** is needed to isolate their effect.

To capture this, I introduce the **MAX IVOL Ratio**, which measures how concentrated the total monthly volatility is in the single day with the largest idiosyncratic return shock. Let $d_{i,t}^*$ denote the day in month t when stock i experiences the maximum absolute return, and let ε_{i,d^*} be the corresponding daily residual from the Fama-French regression. Then, define the squared residual on the MAX day as:

$$R_{i,t}^{MAX} = \varepsilon_{i,d^*}^2 \quad (17)$$

Next, I calculate a version of monthly idiosyncratic volatility that excludes this day, referred to as MAX-adjusted IVOL:

$$IVOL_{i,t}^{MAX_Adj} = \sqrt{\sum_{d \notin d_{i,t}^*} \frac{\varepsilon_{i,t}^2}{n_{i,t} - 2}} \quad (8)$$

Finally, the **MAX IVOL Ratio** is defined as the proportion of total idiosyncratic variance attributable to the most extreme return day:

$$MAX_IVOL_RATIO_{i,t} = \frac{R_{i,t}^{MAX}}{(IVOL_{i,t}^{MAX_Adj})^2} \quad (18)$$

This ratio captures the **concentration** of volatility in a single event. A high value indicates that most of the observed monthly volatility can be attributed to a single outlier, whereas a low value suggests a more evenly distributed volatility profile. Importantly, the ratio is scale-invariant and allows meaningful comparisons across stocks with different overall volatility levels.

This measure is conceptually tied to the literature on lottery-like stocks (Bali, Cakici, and Whitelaw, 2011; Green and Hwang, 2012). While prior work shows that high MAX stocks underperform, it remains unclear whether the **structure** of their volatility, namely, how episodic it is, drives this effect. By sorting portfolios on the MAX IVOL Ratio, I investigate whether the IVOL puzzle is particularly acute in stocks whose volatility is driven by rare, extreme events. If investors ignore or misprice such isolated spikes, we should observe that the IVOL puzzle **weakens or disappears** in high-ratio portfolios, whereas stocks with low MAX IVOL Ratios (indicating diffuse, persistent volatility) continue to exhibit the anomaly.

This diagnostic contributes a new angle to the volatility-pricing literature. Rather than focusing solely on magnitude, it allows for differentiation between **transitory, event-like volatility** and **sustained idiosyncratic risk**, enabling a structural reinterpretation of the conditions under which the IVOL puzzle arises.

3.4 Conceptual Roadmap

This paper addresses the idiosyncratic volatility (IVOL) puzzle by building a multi-layered empirical strategy designed to test three conceptually distinct explanations: **forecast error**, **informational versus non-informational volatility**, and **the structure of volatility**. While much of the prior literature focuses on the negative relationship between realized IVOL and subsequent

returns as evidence of investor mispricing or behavioral bias (e.g., Ang et al., 2006; Bali et al., 2011; Kumar, 2009), this study advances the literature by offering a structured empirical framework that separates measurement from interpretation. Each component of the methodology is designed to isolate a different channel through which the IVOL puzzle may emerge and persist.

The first component investigates the role of **forecast error**. The standard assumption in empirical asset pricing is that volatility is measured ex post, yet asset prices are set based on ex ante expectations. When realized IVOL is used as a proxy for expected risk, empirical tests may conflate **volatility itself with the error in forecasting it**. Bekaert, Bergbrant, and Kassa (2025) offer compelling evidence that the IVOL puzzle is significantly attenuated or disappears altogether when expected IVOL is proxied using forecasts from time-series models, such as ARMA(1,1), rather than using naïve or realized values. They show that the martingale model frequently overestimates future IVOL, especially for illiquid stocks and those exhibiting extreme one-day returns, and that these overestimations disproportionately drive the negative return–IVOL relationship.

To test whether mismeasured risk expectations account for the puzzle, I construct six separate IVOL forecasting models, five based on time-series dynamics (with and without event-based or extreme-return adjustments), and one based on cross-sectional firm characteristics (similar to Gu, Kelly, and Xiu, 2020; Fu, 2009). If the anomaly weakens under better forecasts, this would support the interpretation that the puzzle is not rooted in risk aversion or behavioral bias, but instead in forecast inaccuracy and overreaction to transitory noise.

The second conceptual channel focuses on the **source of volatility**. Volatility may not be homogeneous in its informational content. Events such as earnings announcements are scheduled, widely anticipated, and accompanied by public information releases. Empirical evidence from Beaver (1968), Ball and Shivakumar (2008), and Hirshleifer, Lim, and Teoh (2009) shows that these periods account for large portions of firm-level return variance. Yet, if the volatility associated with such events is predictable and expected, it may not be priced in the same way as volatility that arises on non-event days, such as those influenced by sentiment, rumors, or transient liquidity shocks. This distinction is particularly relevant in light of studies showing that volatility clustering is not always associated with changes in fundamentals (French, Schwert, and Stambaugh, 1987).

To capture this dimension, I decompose realized IVOL into event-driven (EAD) and non-event-driven (NOEAD) components by excluding returns within ± 2 , ± 3 , and ± 5 days around earnings announcements. By constructing separate return-predictive tests for these decomposed volatility measures, I test whether only non-informational volatility is penalized by the market. A finding that the IVOL puzzle persists only for NOEAD volatility would suggest that investors distinguish between fundamental and non-fundamental risk, aligning with theories of limited attention and rational inattention (Sims, 2003; Peng and Xiong, 2006).

The third component addresses the **structure of volatility**. Volatility can result from consistent daily fluctuations or from isolated extreme events, and these structural differences may be critical in understanding how volatility is priced. Stocks with lottery-like features, low price, high volatility, and large right skew, are known to attract retail investor demand and exhibit low subsequent returns (Bali, Cakici, and Whitelaw, 2011; Kumar, 2009). However, prior research has not disentangled whether such anomalies are driven by the **magnitude** of volatility or by its **distributional concentration**. A stock with high IVOL due to one extreme return day may not carry the same pricing implications as a stock with persistent daily fluctuations.

To address this, I develop a novel structural proxy: the **MAX IVOL Ratio**, which captures the share of a firm's monthly volatility attributable to its single most extreme return day. This ratio complements earlier work on the MAX effect by Bali et al. (2011), but shifts the focus from raw maximum returns to volatility composition. The MAX IVOL Ratio quantifies the degree to which monthly volatility is episodic (concentrated) versus diffuse (persistent). If pricing anomalies are stronger in stocks with low ratios, i.e., where volatility is more sustained, this would support the hypothesis that the IVOL puzzle reflects persistent pricing of noise rather than overreaction to rare events.

Taken together, these three elements, forecasting, decomposition, and structure, offer a unified diagnostic framework for identifying the sources of the IVOL puzzle. The forecasting models test whether **misestimated expectations** explain the anomaly; the EAD/NOEAD decomposition tests whether **volatility's informational content** matters for pricing; and the MAX IVOL Ratio tests whether **volatility structure** (episodic vs. persistent) is a key pricing determinant. These components jointly allow me to assess whether the IVOL puzzle represents a true challenge to

rational asset pricing or is the artifact of measurement, event timing, and volatility concentration. The remainder of the paper implements this framework empirically.

4. Data and Variable Construction

This section outlines the construction of the dataset, the variables used in the empirical analysis, and descriptive evidence on the forecast performance of alternative models for expected idiosyncratic volatility (IVOL). The analysis draws on standard data sources widely used in asset pricing and volatility research, and builds an extensive panel of daily and monthly firm-level observations from 1962 to 2023.

4.1 Sample Description and Data Sources

The primary dataset is constructed from the Center for Research in Security Prices (CRSP) and Compustat databases, accessed via WRDS. CRSP provides daily and monthly data on stock returns, prices, volume, and market capitalization. I restrict attention to common stocks (share codes 10 and 11) listed on the NYSE, AMEX, and NASDAQ exchanges. The sample spans January 1962 through December 2023. Stocks are required to have at least 12 daily return observations in a month to be included in the monthly IVOL computation.

Daily factor returns (market excess return, SMB, HML, and RF) are obtained from the Kenneth French data library. To compute residuals for IVOL estimation, I run stock-level Fama-French three-factor regressions on daily returns each month, using the contemporaneous factor returns and requiring a minimum of 12 trading days. These regressions yield the daily idiosyncratic residuals used for constructing realized volatility.

Earnings announcement dates are collected from Compustat's quarterly file. For each firm and month, I define event windows around earnings announcements, including ± 2 , ± 3 , and ± 5 trading days. These are used to separate event-driven volatility (EAD) from non-event-driven volatility (NOEAD) in the construction of several IVOL variations.

Daily and monthly returns are adjusted for delistings following Shumway (1997). Monthly returns are used for portfolio formation and cross-sectional regressions, while daily returns serve as the foundation for volatility measures and diagnostics.

4.3 Summary Statistics: Table 1

Table 1 presents summary statistics for the full panel of firm-month observations, sorted into quintiles based on the realized IVOL in month t . For each quintile, I compute the monthly cross-sectional median of key firm characteristics and IVOL variants. The data span the period from January 1962 through December 2023, and each row in the table reflects the median of the monthly averages across 800 stocks per quintile.

As expected, there is a strong negative relationship between firm size and IVOL. Firms in the lowest IVOL quintile have median log size of 5.75, while those in the highest quintile average only 3.06. This inverse size-volatility relationship is consistent with standard asset pricing results and suggests that higher IVOL stocks tend to be smaller and likely less liquid. Beta increases from 0.45 in the lowest quintile to 1.02 in the highest, indicating that high IVOL firms also exhibit greater systematic risk.

The MAX variable, representing the largest daily return in a month, is strongly increasing across quintiles, from 0.02 in the lowest to 0.14 in the highest, highlighting the episodic nature of volatility in high-IVOL stocks. Momentum, measured over the prior 11 months, declines substantially with IVOL: the lowest IVOL quintile has a median momentum of 10.75%, while the highest quintile exhibits a sharp reversal with -14.90%, consistent with lottery-like stock behavior (Bali et al., 2011; Kumar, 2009).

Illiquidity, as proxied by the Amihud ratio, increases monotonically with IVOL, with a near-zero value in the lowest quintile and 0.06 in the highest. This further supports the view that IVOL is correlated with trading frictions. The volatility components based on EA-adjusted and MAX-adjusted definitions (EA_2, EA_3, EA_5, maxadj) also show upward progression across quintiles, reflecting greater exposure to both scheduled and unscheduled return spikes in high-IVOL firms.

Interestingly, forecasted IVOLs (e.g., IVOL_F, EA_2_F, EA_3_F, etc.) track their realized counterparts closely across ranks, providing preliminary validation for the quality of the forecasting models.

4.4 Forecast Performance: Tables 2 and 3

Tables 2 and 3 present a comprehensive assessment of the out-of-sample forecast accuracy of six IVOL forecasting models under both volatility definitions: standard deviation of daily residuals and the annualized root of squared residuals. These results provide critical insight into which approaches better capture the dynamics of idiosyncratic volatility across firms and time.

In Table 2, Panel A shows that under the standard deviation definition, the **cross-sectional model (IVOL_PREDICT_C)** consistently outperforms the others in terms of both pointwise accuracy and frequency of superiority. It achieves the **lowest average RMSE (0.0216)** and is the best-performing model in **approximately 36.8% of firm-months**, suggesting that firm-level characteristics encode valuable information about volatility persistence. Although the baseline ARMA(1,1) model performs relatively well in mean RMSE and median RMSE, it is clearly dominated in terms of outperformance frequency and robustness across the sample.

The time-series models that adjust for earnings announcements (EA_2, EA_3, EA_5) or exclude extreme return days (MAX-adjusted) perform worse in average rank, indicating that removing data, while potentially beneficial for noise filtering, may also strip away relevant volatility signals. Nevertheless, these filtered models exhibit **modest improvements over the worst-performing baseline**, suggesting that their benefits may be stock-specific or conditional on firm characteristics such as illiquidity or high return skewness.

Panel B of Table 2, based on the RSS definition of IVOL, mirrors these findings. The cross-sectional model again delivers the **lowest RMSE (0.335)** and leads in proportion of best forecasts (36.5%), reinforcing its robustness across volatility definitions. The baseline ARMA(1,1) and MAX-adjusted models also maintain relatively strong performance. The EA-filtered variants lag in rank and accuracy but still demonstrate value relative to naïve forecasts, especially for firms that are prone to event-driven spikes.

Table 3 offers a distributional perspective by reporting **percentiles of the relative RMSE increase** over the best-performing model in each firm-month. The cross-sectional model consistently yields the **lowest median and upper-tail losses**, indicating that its performance is not only strong on average but also highly stable across the distribution. Interestingly, the MAX-adjusted model shows lower RMSE spreads at higher percentiles compared to EA-adjusted forecasts, suggesting that episodic return shocks (such as those driving MAX) are more consequential for forecasting errors than earnings-specific volatility.

These patterns highlight a key contribution of the forecasting framework: while traditional time-series models can perform adequately, especially in low-volatility firms, incorporating structural diagnostics (e.g., MAX effects) and firm characteristics substantially enhances forecast precision. This finding is consistent with recent literature emphasizing the predictive power of cross-sectional firm attributes for volatility and risk modeling (e.g., Gu, Kelly, and Xiu, 2020; Han and Lesmond, 2017). It also suggests that the IVOL puzzle may, in part, reflect **forecasting biases** rooted in model misspecification or information omission.

Taken together, the data construction, variable design, and forecasting evaluation provide a comprehensive platform for examining the pricing implications of idiosyncratic volatility. With multiple forecast models and decomposed IVOL components in hand, the empirical analysis that follows investigates whether the IVOL puzzle persists when expectations are properly accounted for, and whether the pricing of volatility varies by information content and structural composition. The next section turns to asset pricing tests, including portfolio sorts and Fama-MacBeth regressions, to assess the return implications of both realized and forecasted idiosyncratic risk.

5. Return Predictability and the IVOL Puzzle

This section presents the core empirical results linking idiosyncratic volatility (IVOL) to expected stock returns. While standard asset pricing theory predicts a nonnegative relation between risk and expected return, a growing body of empirical evidence challenges this view in the context of firm-specific volatility. Notably, Ang et al. (2006) and Bali and Cakici (2008) document that stocks with high idiosyncratic volatility tend to earn significantly lower future returns, a finding often referred to as the “IVOL puzzle.” This section tests the persistence of the puzzle using an extended sample covering more than six decades (1962–2023), and expands upon prior work by evaluating whether the anomaly remains when using multiple model-based forecasts of expected IVOL rather than realized volatility. We begin with portfolio sorts and Fama-MacBeth regressions based on realized IVOL, and then examine whether the anomaly weakens, disappears, or reverses when replacing ex post volatility with ex ante forecasts derived from ARMA models and cross-sectional prediction frameworks.

5.1 Baseline Results: Realized IVOL and the Puzzle

We begin by evaluating whether the idiosyncratic volatility (IVOL) puzzle documented in prior literature persists in our extended sample of U.S. stocks from January 1962 through December 2023. Specifically, we examine whether stocks with higher realized IVOL in month t earn lower average returns in month $t+1$, contrary to the predictions of standard risk-return tradeoff theories. Consistent with Ang et al. (2006), Bali et al. (2011), and others, we implement two standard asset pricing tests: (1) single-sorted portfolios and (2) monthly Fama-MacBeth regressions.

Table 3 reports equal-weighted (Panel A) and value-weighted (Panel B) returns on decile portfolios formed by sorting stocks each month on realized IVOL, measured as the standard deviation of daily residuals from Fama-French three-factor regressions. In both panels, the first column (IVOL_R) reflects the portfolio-level average returns for each decile under the realized IVOL sorting approach.

The results clearly confirm the existence of the IVOL puzzle. In the equal-weighted portfolios, monthly returns decline from 1.09% in the lowest-IVOL decile to just 0.82% in the highest, resulting in a return spread of -0.27 percentage points. This spread becomes much more pronounced under value weighting, where low-IVOL stocks earn 0.97% while high-IVOL stocks generate a return of -0.37% . The return spread of -1.34% per month is economically large and statistically significant with a t -statistic of -3.92 . These findings align closely with those in Ang et al. (2006), confirming that in a broad cross-section of U.S. equities, higher firm-specific volatility is associated with lower subsequent returns.

To evaluate whether these differences reflect mispricing rather than risk premiums, we also estimate portfolio-level alphas using two factor models. The first is the Fama-French six-factor model (including momentum), and the second is the q -factor model of Hou, Xue, and Zhang (2015). Under both specifications, we find that high-IVOL portfolios generate significantly negative alphas, reinforcing the anomaly. For instance, in the value-weighted panel, the alpha difference between decile 10 and decile 1 under the Fama-French six-factor model is -0.21% ($t = -1.44$), and under the q -factor model it is -0.25% ($t = -1.51$). These results suggest that the negative relation between realized IVOL and returns is not subsumed by known risk factors.

We complement the portfolio results with stock-level regressions in Table 9, Panel A, where we estimate monthly Fama-MacBeth regressions of future excess returns on realized IVOL. The coefficient on IVOL_R is -11.59 ($t = -2.12$), indicating that a one-standard-deviation increase in

realized idiosyncratic volatility is associated with an economically and statistically significant reduction in next month's return. The constant term in the regression is positive and significant (1.39, $t = 7.77$), suggesting that even after accounting for IVOL, the average return remains positive.

Together, these results reaffirm that the IVOL puzzle is a robust empirical regularity. Stocks with high idiosyncratic volatility earn significantly lower returns, particularly in value-weighted portfolios, and this effect persists even after controlling for risk factors or estimation techniques. These patterns suggest that realized IVOL contains pricing-relevant information that is not easily reconciled with standard asset pricing models.

However, the use of ex post realized IVOL in these tests may be problematic. Asset prices are forward-looking, and investors base their risk assessments on expectations of future volatility, not ex post realizations. If realized IVOL is a noisy proxy for expected volatility, particularly for stocks with episodic return spikes or poor liquidity, then the observed return penalty may reflect **forecast error** rather than genuine pricing of risk. In the next section, we explore this possibility by evaluating whether the puzzle persists when expected IVOL is measured using model-based forecasts.

5.2 Forecasted IVOL and the Puzzle

The results from Section 5.1 reaffirm the empirical puzzle that stocks with higher realized idiosyncratic volatility earn lower average future returns, particularly under value weighting. However, these tests rely on ex post volatility, which may not reflect investors' actual expectations at the time of pricing. If the IVOL puzzle is driven not by true risk aversion but by **forecast error**, specifically, overestimation of future volatility, then the anomaly may weaken or vanish when using **ex ante IVOL forecasts**. In this section, I examine whether the return–IVOL relation persists when expected volatility is measured using various forecasting models.

Table 3 includes columns for six forecasted IVOL measures: an ARMA(1,1) time-series forecast (IVOL_F), a cross-sectional predictive model based on firm characteristics (IVOL_C), three earnings-announcement-adjusted forecasts that remove residuals around event windows of ± 2 , ± 3 , and ± 5 days (IVOL_2, IVOL_3, IVOL_5), and a MAX-adjusted forecast (IVOL_MAX) that removes the influence of the maximum daily return in each month. These specifications are

motivated by recent work such as Bekaert et al. (2025), which emphasizes the importance of modeling volatility dynamics and filtering out episodic noise when constructing expected risk measures.

In the equal-weighted portfolios (Table 3, Panel A), the strong negative return pattern observed in realized IVOL (column 1) is no longer present under the forecasted IVOLs. For instance, when sorting on the ARMA(1,1)-based forecast (IVOL_F), average monthly returns increase from 1.12% in the lowest decile to 1.28% in the highest. The return spread is now positive (0.16%), although it is not statistically significant. The cross-sectional forecast (IVOL_C) shows a similar pattern, with a return spread of 0.23% between the high and low deciles. These results indicate that, at least under equal weighting, the IVOL puzzle largely disappears when volatility expectations are modeled appropriately.

The value-weighted portfolios in Panel B yield more nuanced insights. While the negative return spread of -1.34% for realized IVOL (IVOL_R) is large and statistically significant ($t = -3.92$), the return spreads for all six forecasted IVOL models also remain **statistically significant**, though the economic magnitudes are generally smaller. For example, sorting on the ARMA(1,1)-based forecast (IVOL_F) yields a return spread of -0.86% with a t-statistic of -2.21 , while the cross-sectional forecast (IVOL_C) generates a spread of -1.22% ($t = -3.95$). Similar patterns hold for the earnings announcement-adjusted and MAX-adjusted forecasts, where spreads range from -0.77% to -0.84% , with t-statistics between -2.09 and -2.27 . These findings suggest that the IVOL puzzle **persists in value-weighted portfolios even when expected IVOL is modeled**, although the anomaly weakens somewhat relative to realized IVOL.

These results offer partial support for the hypothesis that the IVOL puzzle is driven, in part, by measurement error, particularly instances where realized volatility substantially overstates expected volatility. In equal-weighted portfolios, where smaller and less liquid stocks dominate, the anomaly **largely disappears** when expected IVOL is modeled using time-series forecasts or firm-level characteristics. This suggests that much of the anomaly in these stocks may reflect temporary pricing noise or volatility mismeasurement. However, in value-weighted portfolios, where larger stocks carry more influence, the puzzle **persists across all six forecasted IVOL specifications**, with return spreads remaining economically and statistically significant. These findings point to a more complex picture: while improved forecasts reduce the severity of the

anomaly, they do not fully eliminate it, especially in the broader market. This evidence suggests that the puzzle cannot be attributed solely to forecast error and highlights the need to examine the **source and structure of volatility**, which we explore further in the next sections.

The Fama-MacBeth regressions in Table 4 further corroborate these findings. When regressing future excess returns on realized IVOL (IVOL_R), the coefficient is -11.59 ($t = -2.12$), consistent with the strong anomaly documented in the portfolio analysis. However, when using the ARMA(1,1)-based forecast (IVOL_F), the coefficient shrinks dramatically to -2.80 and becomes statistically insignificant ($t = -0.32$). Similar attenuation is observed across the remaining forecasted IVOL measures, with coefficients ranging from -1.75 (MAX-adjusted) to -5.54 (cross-sectional), all of which are statistically indistinguishable from zero.

These regression results imply that once we account for investor expectations more accurately, IVOL no longer predicts negative future returns. The combination of attenuated coefficients and diminished portfolio return spreads under forecasted volatility suggests that the IVOL puzzle may be less a reflection of mispriced risk and more a byproduct of forecast error and volatility shocks. Interestingly, the cross-sectional forecast model (IVOL_C), which integrates firm-level attributes such as MAX, MOM, size, and illiquidity, produces the most economically intuitive results, flat or weakly positive return patterns and insignificant pricing coefficients, highlighting the benefit of incorporating firm-specific information into risk forecasts.

In sum, the evidence from both portfolio sorts and Fama-MacBeth regressions suggests a more nuanced view of the IVOL puzzle. When expectations of idiosyncratic volatility are modeled using ARMA time-series techniques, cross-sectional predictors, and volatility filtering, the negative return–volatility relation **largely dissipates in equal-weighted portfolios**, where small, illiquid firms dominate. However, in **value-weighted portfolios**, the anomaly **persists across all six forecasting methods**, with high-IVOL stocks continuing to underperform significantly. This indicates that while forecast error may explain part of the puzzle, particularly for smaller firms, the anomaly cannot be fully dismissed as a statistical artifact. Instead, the results point to the importance of understanding **what drives idiosyncratic volatility**, not just how well it is forecasted. In the next section, I explore this further by decomposing realized IVOL into event-driven and non-event-driven components to test whether the **source** of volatility has implications for pricing.

5.3 Decomposition: EAD vs. NOEAD IVOL

A natural question arising from the IVOL puzzle is whether all sources of idiosyncratic volatility are treated equally by investors. Specifically, do investors price volatility related to earnings announcements (which is often associated with fundamental information flow), differently from volatility unrelated to such events? To address this question, I decompose total realized IVOL into two components: volatility that occurs within the earnings announcement window (EAD-IVOL), and volatility that occurs outside that window (NOEAD-IVOL).

For each firm-month, EAD-IVOL is defined as the standard deviation of residuals (from the Fama-French three-factor model) computed using only those trading days that fall within symmetric event windows around the earnings announcement date. I construct three windows: ± 2 , ± 3 , and ± 5 trading days. If a firm does not have an earnings announcement in a given month, its EAD-IVOL is coded as zero. The complementary component, NOEAD-IVOL, is computed as the standard deviation of daily residuals outside these windows, again using the same FF3 regression framework. For comparison, I also compute EAD and NOEAD volatility by excluding the maximum return day from each month, labeled EAD_MAX and NOEAD_MAX, respectively.

Table 6 reports the results of monthly Fama-MacBeth regressions where the dependent variable is next month's return and the independent variable is either EAD-IVOL or NOEAD-IVOL (based on the ± 2 , ± 3 , ± 5 windows and MAX exclusion). The intercepts across all regressions remain stable and positive, ranging from 1.16 to 1.37 and highly statistically significant, suggesting positive average returns across the sample. The more striking variation lies in the coefficients on the volatility components.

Starting with the EAD-IVOL regressions, we see no evidence of a negative return–volatility relationship. The coefficient on EAD_2 is slightly negative (-0.28) but statistically insignificant ($t = -0.14$), while EAD_3 and EAD_5 both exhibit positive coefficients (1.18 and 1.00) with t -statistics of 0.57 and 0.46, respectively. These results suggest that **investors do not penalize firms for earnings-related volatility**; in fact, there is some weak evidence that such volatility may be positively associated with future returns, potentially because it reflects the arrival of firm-specific information.

In sharp contrast, the NOEAD-IVOL components exhibit strong and statistically significant negative coefficients. For example, NOEAD_2 yields a coefficient of -12.57 ($t = -2.26$), and

similar magnitudes are found for NOEAD_3 (-12.64 , $t = -2.28$) and NOEAD_5 (-12.51 , $t = -2.27$). Even when the maximum return day is excluded (NOEAD_MAX), the coefficient remains highly negative (-11.98 , $t = -1.64$). These findings indicate that volatility arising outside of earnings events is consistently associated with lower expected returns, supporting the hypothesis that investors view such volatility as noise or pricing inefficiency rather than fundamental risk.

The asymmetric pricing of EAD versus NOEAD volatility contributes a novel insight to the literature on the IVOL puzzle. It suggests that the puzzle is not simply about the level of volatility, but about its **source**. While earnings-related volatility may convey valuable information about firm fundamentals, leading investors to discount it less or even reward it, non-event-driven volatility appears to be interpreted as non-informative and is therefore penalized. This interpretation is consistent with behavioral theories of mispricing, where investors overreact to salient but non-fundamental signals (Barberis, Shleifer, and Vishny, 1998), as well as with rational models that predict no compensation for diversifiable noise (Merton, 1987).

These findings also complement the earlier forecasting results. In particular, they suggest that forecasts which filter out EAD-related volatility (as in the EA-adjusted ARMA models) may reduce the puzzle in some specifications because they better align with how investors distinguish between meaningful and non-meaningful volatility. More broadly, this decomposition strengthens the argument that the IVOL puzzle is not uniform across types of volatility, a key point that I build on in the next section, which investigates the structural nature of volatility using the MAX IVOL Ratio diagnostic.

5.3.1 Double-Sorted Portfolios: MAX and Realized IVOL

To further investigate the interaction between extreme return behavior and idiosyncratic volatility, I sort stocks into portfolios based on both their maximum daily return in month t (MAX) and their realized idiosyncratic volatility in the same month. Table 7 presents the results from 5×5 double sorts on MAX (rows) and realized IVOL (columns), reporting the average returns of the resulting portfolios in month $t+1$, as well as associated alphas from Fama-French 6-factor (FF6) and quality-minus-junk (Q-factor) models.

In both equal-weighted and value-weighted sorts, a consistent pattern emerges: the negative relation between IVOL and next month returns, the IVOL puzzle, is concentrated in the highest

MAX quintiles. For example, in Panel A (equal-weighted), the return spread between high- and low-IVOL stocks within the high-MAX group is -0.11% , with statistically significant negative FF6 and Q-factor alphas (-0.65 and -0.60 , respectively). In Panel B (value-weighted), the spread is more pronounced: moving from the lowest to highest IVOL quintile within the highest MAX group, average returns decline sharply from 0.64% to -1.12% , with an extreme return spread of -1.76% ($t = -5.09$) and highly significant negative alphas.

Conversely, in low-MAX portfolios, the IVOL puzzle either disappears or reverses. In the equal-weighted case, the return spread from low to high IVOL in the lowest MAX quintile is $+0.69\%$ ($t = 5.17$), and both FF6 and Q-factor alphas are significantly positive. In value-weighted portfolios, returns across IVOL quintiles within the low-MAX group are flat or slightly positive, with no significant evidence of a puzzle.

These results indicate that the pricing of idiosyncratic volatility depends critically on the return distribution's tail structure. Specifically, the puzzle is strongest when IVOL is paired with extreme one-day spikes, consistent with prior literature linking MAX to investor preferences for lottery-like stocks (Bali, Cakici, and Whitelaw, 2011; Green and Hwang, 2012). In these cases, high-IVOL stocks tend to exhibit speculative behavior and pricing distortions, which may not reflect rational compensation for risk.

The sharp contrast between the high-MAX and low-MAX groups suggests that not all volatility is priced equally, and that the structure of volatility may be more important than its level. This motivates the development of a new diagnostic measure that captures whether a firm's volatility is episodic (spike-driven) or diffuse (persistent). In the next section, I introduce the MAX IVOL Ratio as a proxy for volatility structure, and explore how this ratio helps explain the cross-sectional variation in returns that conventional IVOL measures cannot.

5.4 Diagnostic Sorting: MAX IVOL Ratio and the Structure of Volatility

While most studies treat idiosyncratic volatility as a scalar measure of risk, this approach implicitly assumes that all volatility is distributed uniformly throughout the return process. However, this assumption may not hold in reality, particularly in the presence of **extreme one-day return events**. When volatility is concentrated in rare, episodic spikes, its economic interpretation may

shift from a signal of risk to a reflection of noise, liquidity shocks, or speculative pricing. This insight motivates the development of a new diagnostic measure: the MAX IVOL Ratio.

Formally, I define the MAX IVOL Ratio for firm i in month t as:

$$MAX_IVOL_RATIO_{i,t} = \frac{R_{i,t}^{MAX}}{(IVOL_{i,t}^{MAX_Adj})^2} \quad (18)$$

where $R_{i,t}^{MAX}$ is the residual squared from the Fama-French three-factor model on the day of the firm's maximum daily return, and $(IVOL_{i,t}^{MAX_Adj})^2$ is the total monthly residual variance computed from all other trading days (i.e., excluding the MAX day). This ratio captures the proportion of monthly idiosyncratic volatility attributable to a single extreme return, effectively distinguishing between episodic and diffuse volatility structures. A high ratio implies volatility is concentrated in rare events, while a low ratio indicates more evenly distributed noise.

To test whether this volatility structure matters for return pricing, I sort firms into decile portfolios each month based on their MAX IVOL Ratio, and report equal- and value-weighted returns in Table 8. Several notable patterns emerge.

In the equal-weighted portfolios, I observe a monotonic decline in average returns from low- to high-ratio deciles. The lowest ratio group (Low_F) earns 1.15% per month, while the highest ratio group (High_F) earns just 0.87%, resulting in a spread of -0.29% ($t = -3.10$). This return differential is echoed in the factor-adjusted alphas: FF6 alpha = -0.33% ($t = -3.83$), Q-factor alpha = -0.30% ($t = -3.16$). These results are both economically meaningful and statistically significant, and suggest that when volatility is diffuse, investors are less likely to penalize it, but when it is concentrated in a single day, investors assign a valuation discount.

In the value-weighted portfolios, the pattern is less pronounced. While returns do not decline monotonically, the return spread between extreme portfolios is $+0.07\%$ ($t = 0.65$), statistically insignificant. However, the FF6 and Q-factor alphas remain significantly negative in the highest ratio group (-0.29% and -0.32% , both with t-stats above -2.75), suggesting that even among larger firms, concentrated idiosyncratic spikes are penalized in risk-adjusted terms.

Taken together, these findings reinforce the hypothesis that how volatility is structured, spiky versus persistent, is a critical determinant of its pricing. The IVOL puzzle, as observed in earlier sections, may not stem from volatility per se, but from the episodic nature of certain extreme return

days, which generate high realized volatility but carry limited informational content. These high-ratio firms may represent speculative sentiment, liquidity imbalances, or transient shocks, all of which are typically not rewarded by risk-averse investors.

This interpretation dovetails with existing literature on lottery preferences and episodic mispricing. Bali, Cakici, and Whitelaw (2011) argue that investors may irrationally favor stocks with lottery-like return distributions (e.g., high MAX), leading to overpricing and future underperformance. Green and Hwang (2012) similarly show that stocks with a history of sharp return spikes exhibit abnormally low risk-adjusted returns. My MAX IVOL Ratio extends this logic by distinguishing between stocks where such spikes dominate the return variance, versus those where volatility is more stable and potentially information-based.

Ultimately, these results highlight that the structure of idiosyncratic volatility is economically informative. Conditioning on the MAX IVOL Ratio offers a powerful diagnostic tool for identifying when high IVOL reflects legitimate risk versus when it masks episodic noise or sentiment-driven distortions. In the next section, I bring together these findings in a unified framework, showing how forecast errors, volatility decomposition, and structure jointly shape the pricing of IVOL in the cross section of returns.

5.5 Putting It All Together

The analyses thus far have demonstrated that the IVOL puzzle is not a monolithic phenomenon, but one that is highly conditional on the nature and quality of volatility forecasts. In this section, I synthesize the core findings from earlier parts of the study by drawing on five complementary sets of portfolio analyses. Together, these tests build a coherent picture that reveals the mechanisms underlying the puzzle and isolates its economic drivers. Specifically, I use double-sorted portfolios and forecast error partitions to evaluate how episodic noise, measurement error, and volatility structure interact with future returns.

MAX $\times$ IVOL Portfolios: Volatility Puzzle Concentrated in High-MAX Stocks

The first double-sort partitions stocks by MAX, the maximum daily return in a given month, and, within each MAX quintile, further sorts by realized IVOL. As shown in Table 7, the IVOL puzzle is not uniform across MAX groups. In equal-weighted portfolios, the relation between IVOL and

future returns is largely flat or even positive in low-MAX groups but turns significantly negative in high-MAX groups. For instance, in the highest MAX quintile, high-IVOL portfolios underperform low-IVOL portfolios by -0.98% per month, with a FF6 alpha of -1.43% and a Q-factor alpha of -1.38%, both statistically significant.

This pattern becomes even stronger in value-weighted portfolios, where the spread between high- and low-IVOL portfolios within the high-MAX quintile reaches -2.16% ($t = -5.42$), with alphas of -2.45% and -2.40% under the FF6 and Q-factor models, respectively. These results suggest that the IVOL puzzle is concentrated in stocks exhibiting extreme single-day returns. Such return spikes, which may be driven by illiquidity, market frictions, or attention-grabbing news, distort the measurement of volatility and inflate perceived risk. This evidence aligns with Bali et al. (2011) and Green and Hwang (2012), who emphasize that investors may overweight such extreme events due to lottery-like preferences or behavioral biases.

MAX $\times$ MAX IVOL Ratio Portfolios: Volatility Structure as a Key Driver

To further explore the role of episodic noise, I construct the **MAX IVOL Ratio**, defined as the squared residual of the FF3 model on the maximum return day divided by the monthly MAX-adjusted IVOL squared. This ratio proxies for the share of total idiosyncratic volatility explained by a single day's spike. Table 9 presents double-sorted portfolios based on MAX and this ratio.

In value-weighted portfolios, the results are striking: within each MAX group, high-ratio stocks consistently deliver higher returns. For example, in the high-MAX quintile, the high-ratio portfolio earns 0.54% per month versus 0.10% for the low-ratio group, yielding a spread of +0.44%. Although the alphas are slightly negative, they are much less pronounced than in the previous sort. The equal-weighted results also show a similar monotonic pattern, with high-ratio stocks outperforming low-ratio ones. These findings suggest that episodic, spiky volatility may actually carry a positive premium, possibly because it reflects event-related uncertainty or time-varying risk.

This analysis reframes the puzzle: rather than IVOL being uniformly mispriced, it appears that the structure of volatility matters. When idiosyncratic risk is concentrated in one extreme day, it may convey different information, or noise, than when volatility is more diffuse.

Ratio $\times$ IVOL Portfolios: When Noise Dominates, IVOL is Penalized

In Table 10, I present double-sorted portfolios based on the MAX IVOL Ratio and realized IVOL. The results strongly reinforce the argument that the pricing of IVOL is contingent on the nature of volatility. In value-weighted portfolios, the IVOL puzzle is especially strong in the highest ratio quintile. For these stocks, the return spread between high- and low-IVOL portfolios is -0.76%, and the Q-factor alpha is -1.22% ($t = -3.55$). Interestingly, even in equal-weighted portfolios, which generally do not show a strong IVOL puzzle, high-ratio stocks exhibit significantly negative IVOL return spreads.

This asymmetry is crucial: when volatility is structured as “noise” (i.e., dominated by one extreme event), high-IVOL stocks underperform. But when that structure is absent, as in low-ratio stocks, IVOL is not strongly priced. This suggests that investors are sensitive not just to the level of volatility, but to how it is generated. Diffuse volatility, possibly tied to fundamental uncertainty, is treated differently from episodic volatility, which may be perceived as overreaction or market noise.

Forecast Error Partitions: Overestimation Drives the Puzzle

The final dimension of analysis examines whether the puzzle is driven by systematic errors in volatility forecasting. I split the sample into two groups: (1) stocks where the ARMA- or cross-section-based forecast overestimates realized IVOL, and (2) stocks where it underestimates IVOL. Tables 10 and 11 report single-sorted portfolio results for these groups.

The contrast is stark. Among **overestimated IVOL stocks** (Table 11), the IVOL puzzle is amplified: in equal-weighted portfolios, the return spread between high- and low-IVOL stocks is -2.65%, with t-stats exceeding 8 for raw returns and over 6 for both FF6 and Q-factor alphas. Similar patterns appear in value-weighted portfolios, with spreads of -2.33% and t-stats exceeding 7.5. These findings mirror Bekaert et al. (2025), who argue that extreme overpredictions in IVOL, often linked to illiquid or noisy stocks, are the source of the puzzle.

In sharp contrast, **underestimated IVOL stocks** (Table 12) exhibit no puzzle. In fact, high-IVOL portfolios earn significantly **higher** returns. Equal-weighted portfolios show return spreads of +5.15% ($t = 7.17$), with large positive alphas. Even in value-weighted portfolios, high-IVOL stocks outperform low-IVOL ones by +3.22%, again with positive albeit smaller alphas. This inversion of the puzzle supports the view that IVOL may only appear mispriced when forecasts are

systematically too high, potentially due to noisy volatility measures driven by MAX days or low liquidity.

Synthesis: A Unified Interpretation

Taken together, these five empirical tests provide a coherent resolution to the IVOL puzzle. The evidence suggests that:

The puzzle is not pervasive but is largely concentrated in stocks with high MAX returns and in months where IVOL is overestimated relative to its subsequent realization.

The structure of volatility matters. The MAX IVOL Ratio reveals that episodic volatility, where one day dominates total variation, is associated with poor future returns, even conditional on IVOL levels.

When volatility is diffuse or fundamentals-driven, as in low MAX or low ratio stocks, the IVOL-return relation flattens or reverses. These stocks do not exhibit the negative return–volatility link.

Forecast errors are central. The most severe underperformance of high-IVOL stocks occurs when forecasts are too pessimistic. When IVOL is underestimated, high-volatility stocks generate strong future returns.

Volatility is not mispriced, it is misunderstood. Traditional measures of realized IVOL embed too much noise and episodic distortion. More accurate forecasts, adjusted for MAX and EA-driven noise, eliminate much of the puzzle.

By bringing these results together, this section underscores a fundamental insight: the IVOL puzzle is not a true anomaly but a statistical artifact. It arises when investors (or econometric models) misinterpret the informational content of idiosyncratic volatility. Decomposing, forecasting, and structurally analyzing volatility allows for a more precise understanding of its risk-return tradeoffs. This perspective suggests that future research should prioritize not just levels of volatility, but also their origins, dynamics, and temporal concentration.

6. Robustness

To ensure that the results presented in this study are not specific to the full-sample period or overly sensitive to particular data segments, I conduct a series of robustness checks. The most comprehensive of these is a temporal split of the sample into two distinct subperiods: **1962–1993** and **1993–2023**. This split is designed to capture meaningful differences in institutional context, trading behavior, and the evolution of volatility modeling over time. The former period reflects a pre-electronic trading, lower-frequency environment with limited investor sophistication, while the latter encompasses the rise of algorithmic trading, decimalization, and dramatic increases in trading volume, liquidity fragmentation, and investor heterogeneity.

By rerunning all key empirical analyses, including single- and double-sorted portfolio returns, Fama-MacBeth regressions, and diagnostic ratio sorts, I test whether the IVOL puzzle and its resolution through forecasting and decomposition persist across both subsamples. This approach provides a strong test of temporal stability and guards against the possibility that the findings are driven by idiosyncrasies in a particular market regime.

6.1 Persistence of the IVOL Puzzle in the Pre-1993 Era

In the **1962–1993 subperiod**, the classic IVOL puzzle appears robustly across realized volatility measures. Value-weighted portfolios sorted on realized IVOL continue to show a pronounced negative relation with future returns, particularly in the highest decile, consistent with Ang et al. (2006) and Bali and Cakici (2008). Equal-weighted portfolios, in contrast, show a much weaker pattern, reinforcing the notion that the puzzle is concentrated among larger, more liquid firms that dominate the value-weighting schemes.

Crucially, when I apply the full suite of **forecasting models**, including ARMA(1,1), EA-adjusted IVOLs, MAX-adjusted IVOL, and the cross-sectional prediction model, I observe the same attenuation of the puzzle as in the full sample. In equal-weighted portfolios, the negative IVOL-return relation disappears or reverses in nearly every forecasting specification. In value-weighted portfolios, although the IVOL puzzle persists, its magnitude is sharply reduced relative to the baseline realized IVOL sorts. The Fama-MacBeth regressions also support this finding, with the coefficient on realized IVOL being significantly negative, while the coefficients on forecasted IVOLs are smaller and statistically indistinguishable from zero.

This persistence of findings in the earlier era is particularly noteworthy, as it confirms that the attenuation of the puzzle through improved volatility modeling is not a consequence of post-1990s data features, such as high-frequency noise, structural breaks in volatility, or increased cross-sectional return dispersion. Instead, the mechanism underlying the puzzle, and its resolution, appears consistent with a more fundamental property of mismeasured or episodic volatility.

6.2 Replication in the 1993–2023 Era

In the more recent **1993–2023 subperiod**, I find results that are nearly identical in direction and strength. The IVOL puzzle remains present when using realized volatility, especially in value-weighted portfolios, where high-IVOL deciles underperform significantly. Equal-weighted portfolios again show weak or flat patterns.

Forecast-based IVOL models continue to demonstrate an improved risk-return relation, with the puzzle disappearing in equal-weighted portfolios and substantially weakening in value-weighted ones. Importantly, the **cross-sectional prediction model** based on MAX, MOM, size, and ILLIQUIDITY performs well in both subperiods, providing further evidence of its robustness and generalizability.

The decomposition tests, EAD vs. NOEAD, reveal that the positive coefficient on EAD-IVOL and the strongly negative, significant coefficient on NOEAD-IVOL are consistent across both periods. This indicates that the pricing of idiosyncratic volatility is driven not by the existence of high volatility per se, but by its underlying structure and informational content. Volatility surrounding earnings announcements, which tends to reflect information updates, is not penalized by the market; in contrast, background volatility in non-event windows, often driven by noise or liquidity shocks, is met with negative returns.

The same conclusion holds for the **MAX IVOL Ratio** diagnostics. In both subperiods, high-ratio stocks, those where a large share of volatility is explained by a single day, underperform sharply. Portfolios sorted on this ratio exhibit strong monotonic return patterns, especially in value-weighted results, suggesting that the episodic nature of volatility is a key determinant of its pricing implications, regardless of market structure or trading technology.

6.3 Implications of Temporal Robustness

These subperiod findings greatly enhance the credibility of the overall conclusions. First, they demonstrate that the attenuation of the IVOL puzzle through improved forecasting and decomposition is not limited to a particular era or market environment. Whether under the more institutionally stable, low-frequency conditions of the pre-1990s or the technologically dynamic post-1990s period, the core result holds: **once idiosyncratic volatility is measured or forecasted more accurately, its apparent mispricing diminishes or disappears.**

Second, the consistency across eras suggests that the puzzle is not driven by evolving preferences, structural market changes, or factor mismeasurement specific to any one time. Instead, it reflects a deeper empirical issue: realized IVOL contains a mix of noise and information, and standard asset pricing models fail when these are not properly separated. By incorporating temporal robustness into the analysis, I demonstrate that this interpretation is not contingent on specific data artifacts or time-specific anomalies.

Finally, the strength of the findings in the 1993–2023 subperiod, which includes periods of extreme market volatility (e.g., the 2000 dot-com crash, 2008 financial crisis, 2020 COVID-19 shock), implies that the diagnostic tools developed in this study, especially the MAX IVOL Ratio, are useful even in stressed or non-normal conditions. This enhances their potential application in practice, as portfolio managers and risk modelers often seek tools that remain effective in crisis periods.

7. Discussion and Implications

This paper set out to reexamine the idiosyncratic volatility (IVOL) puzzle through a forecasting-based lens, motivated by the idea that realized IVOL may poorly approximate investor expectations. Through an extensive battery of predictive models, decomposition techniques, and diagnostic volatility structure tests, I show that much of the puzzle dissipates once expected volatility is measured with greater precision. This section contextualizes these findings and discusses their broader implications for asset pricing theory, investor behavior, and financial modeling.

7.1 Revisiting the Nature of the IVOL Puzzle

At the core of this study is the empirical observation that realized idiosyncratic volatility, often measured as the standard deviation or root-sum-square of residuals from factor models, bears a negative or null relationship with expected stock returns. This violates foundational principles in asset pricing, which predict that higher risk should be compensated with higher return (Sharpe, 1964; Merton, 1987). Consistent with prior work (e.g., Ang et al., 2006; Bali and Cakici, 2008), I confirm this anomaly in value-weighted portfolios and Fama-MacBeth regressions when using realized IVOL. However, I also show that the anomaly weakens or vanishes in equal-weighted portfolios and, more importantly, becomes attenuated or disappears once IVOL is forecasted.

This suggests that the puzzle arises not necessarily because investors misprice risk, but rather because **realized IVOL is a noisy proxy for expected risk**. As in Bekaert et al. (2025), my results show that extreme overpredictions in IVOL are disproportionately responsible for the negative IVOL-return relation. Once these are filtered through ARMA-based forecasts, event-adjusted IVOLs, or cross-sectional prediction models, the puzzle becomes considerably less pronounced. In equal-weighted portfolios, the return–IVOL relation often turns flat or positive. Even in value-weighted portfolios, where larger firms dominate, the negative relation is no longer statistically robust across forecasted volatility measures.

The fundamental takeaway is that volatility matters, but **how** volatility is generated and measured matters more. Episodic volatility spikes that arise from rare events or extreme trading days distort realized IVOL in ways that standard asset pricing models do not capture. Investors appear to price persistent volatility risk more rationally, while penalizing volatility that is transient, noisy, or associated with illiquidity and information frictions.

7.2 Behavioral vs. Rational Interpretations

The findings in this paper contribute to an ongoing debate over whether the IVOL puzzle arises from behavioral mispricing or unmodeled risk factors. While rational explanations, such as diversification constraints or incomplete markets (Merton, 1987; Campbell and Vuolteenaho, 2004), have been proposed, my results offer stronger support for **behavioral interpretations** centered on investor attention, misreaction to extreme events, and noise trading.

Specifically, the role of **overestimated volatility forecasts**, which are shown to be disproportionately associated with high MAX returns, low momentum, and high illiquidity, suggests that investors overreact to idiosyncratic shocks that do not reflect future cash flow

volatility. This aligns with the insights of Barberis and Huang (2008) and Green and Hwang (2012), who argue that investors exhibit preference for lottery-like stocks and overweigh salient but unlikely outcomes.

Further, the pricing asymmetry observed in the EAD vs. NOEAD decomposition reinforces this narrative. Event-related volatility, such as that occurring around earnings announcements, is not penalized by markets and sometimes even priced positively. In contrast, **background volatility in non-event periods**, which lacks fundamental informational content, attracts a return discount. These patterns cannot be reconciled with standard risk-return tradeoffs but are consistent with behavioral pricing errors arising from signal-to-noise confusion.

The MAX IVOL Ratio results strengthen this view by showing that volatility structure, not just volatility level, is priced by investors. Stocks where volatility is concentrated in one or two extreme days perform significantly worse than stocks with diffuse volatility, even when the total IVOL is the same. This suggests that investors are especially averse to noisy, episodic volatility, and may demand higher compensation only for persistent volatility that reflects long-term uncertainty.

7.3 Implications for Asset Pricing Models

The results have important implications for how idiosyncratic risk is modeled and incorporated into empirical asset pricing frameworks. First, they challenge the widespread use of realized IVOL as a proxy for expected risk. As Frazzini and Pedersen (2014) emphasize, the failure to account for frictions or behavioral pricing errors leads to anomalies that appear inconsistent with theory. My results echo this insight: **risk must be forecasted, not merely observed, if it is to align with equilibrium pricing models.**

Second, the study points to a need for greater attention to volatility structure in empirical finance. While traditional multifactor models focus on risk exposures (e.g., beta), my findings suggest that residual variance itself can contain economically meaningful structure that affects returns. By introducing diagnostics such as the MAX IVOL Ratio, I provide tools that can be integrated into factor construction or used to refine risk forecasts, particularly in models that rely on conditional heteroskedasticity (e.g., GARCH-based asset pricing).

Third, the consistent pricing of NOEAD volatility, negative and significant in all regressions, raises the possibility that idiosyncratic noise, rather than undiversifiable risk, may be driving

pricing anomalies. This may motivate future models that explicitly differentiate between **informational and non-informational volatility**, perhaps along the lines of endogenous noise models or models with attention constraints (e.g., Peng and Xiong, 2006).

7.4 Practical Implications

While this study is primarily empirical and academic in nature, it also offers actionable insights for investment professionals. First, portfolio managers who rely on realized IVOL as a risk signal may be misled by transitory volatility spikes. By incorporating **forecasted IVOLs** or filtering out noise from earnings or extreme returns, managers can construct more robust factor portfolios and avoid systematic underperformance associated with volatility-chasing strategies.

Second, the MAX IVOL Ratio offers a simple yet powerful screening tool for identifying stocks whose volatility is driven by ephemeral shocks rather than persistent uncertainty. This has applications in portfolio construction, hedge design, and even options pricing, where distinguishing noise from risk is critical.

7.5 Limitations and Future Directions

Like all empirical studies, this paper has limitations. The analysis is based on U.S. stocks from 1962 to 2023, and while the findings are robust across subperiods, their generalizability to international markets or alternative asset classes remains to be tested. Moreover, the use of ARMA and cross-sectional regressions assumes stationarity and may not capture regime shifts or structural breaks in volatility dynamics.

Future work could extend the forecasting framework to high-frequency data, explore applications to options markets, or test whether volatility structure measures can improve expected return forecasts in machine learning-based models. Another promising avenue is to examine investor flows or trading behavior around extreme volatility episodes, which could provide direct evidence of behavioral mispricing mechanisms.

8. Conclusion

This paper reexamines the idiosyncratic volatility (IVOL) puzzle by focusing on the distinction between realized and expected volatility. While the literature has long documented a negative relationship between IVOL and future stock returns, an apparent violation of traditional risk-return

tradeoffs, this anomaly may largely stem from measurement issues rather than true mispricing. Building on recent work by Bekaert et al. (2025), I show that much of the IVOL puzzle weakens or disappears when volatility is forecasted rather than observed.

Using a comprehensive dataset of CRSP stocks from 1962 to 2023, I implement six distinct IVOL forecasting approaches, including ARMA(1,1) time-series models with raw and adjusted return inputs, and a cross-sectional predictive model based on firm characteristics such as momentum, illiquidity, and extreme return events (MAX). Across both portfolio sorts and Fama-MacBeth regressions, I find that forecasted IVOLs exhibit weaker and often insignificant relationships with future returns, particularly in equal-weighted settings. These findings suggest that the anomaly is more pronounced when volatility is measured ex post and contaminated by non-informational shocks.

The analysis goes further by decomposing realized volatility into earnings-announcement-driven (EAD) and non-event-driven (NOEAD) components. I show that EAD IVOL is not associated with negative return premiums and may even be positively priced, while NOEAD IVOL, interpreted as noise volatility, drives the negative return-IVOL relation. Additionally, I introduce a new diagnostic ratio capturing the squared residual from the single most volatile day relative to the month's volatility excluding that spike. This MAX IVOL Ratio effectively separates stocks with diffuse risk from those with concentrated, episodic volatility. Portfolios sorted on this ratio reveal that investors systematically penalize stocks with spiky, noisy volatility structures, further confirming that the IVOL puzzle is closely linked to measurement distortion.

In sum, this paper offers a new lens through which to view the IVOL anomaly. Rather than being a failure of risk-based asset pricing models, the puzzle appears to reflect systematic overestimation of volatility risk, particularly when driven by rare events or informationally unimportant noise. By refining how volatility is forecasted and decomposed, and by introducing diagnostics that capture volatility structure, this paper contributes both conceptually and methodologically to the asset pricing literature.

Future research can build on these findings by exploring other forms of volatility decomposition, examining market microstructure drivers of noise, or testing the implications of volatility structure in international or derivatives markets. More broadly, the results underscore the need for careful

modeling of investor expectations, an insight that may extend to other asset pricing anomalies beyond IVOL.

Appendix-1

Table 1. Out-of-Sample Forecast Performance for IVOL Models

This table reports the out-of-sample forecast performance of six IVOL prediction models across two definitions of realized volatility. Panel A evaluates forecasts where realized IVOL is computed as the monthly standard deviation of residuals from the Fama-French three-factor model. Panel B uses an alternative measure based on the annualized square root of the sum of squared residuals. Each row presents the model's average root mean squared error (RMSE), the proportion of months in which it delivers the best forecast, its average rank across months, the improvement relative to the worst model, and the median RMSE. Forecast models include ARMA(1,1) applied to time series of IVOL constructed from residuals (IVOL_F), earnings-adjusted variants excluding ± 2 , ± 3 , or ± 5 days (IVOL_2F, IVOL_3F, IVOL_5F), a MAX-adjusted version removing the month's highest residual day (IVOL_maxadjF), and a cross-sectional forecast based on firm characteristics (IVOL_CF).

Panel A: IVOL based on standard deviation of residuals

Model	Mean RMSE	Proportion Best	Average Rank	Improvement to Worst	Median RMSE
IVOL_F	0.022300	0.226664	3.132787	0.610221	0.015285
IVOL_2F	0.022337	0.086090	3.158406	0.610735	0.015279
IVOL_3F	0.022368	0.071936	3.345196	0.596146	0.015302
IVOL_5F	0.022773	0.071834	3.991706	0.473967	0.015528
IVOL_maxadjF	0.022347	0.175680	3.807224	0.493603	0.015187
IVOL_CF	0.021635	0.367796	3.564681	0.512542	0.015185

Panel B: IVOL based on the annualized square root of the sum of squared residuals

Model	Mean RMSE	Proportion Best	Average Rank	Improvement to Worst	Median RMSE
IVOL _s _F	0.34483	0.24254	3.08279	0.62053	0.23634
IVOL _s _2F	0.34548	0.07960	3.12584	0.61800	0.23624
IVOL _s _3F	0.34597	0.06707	3.35451	0.60228	0.23638
IVOL _s _5F	0.35234	0.07148	4.02356	0.47303	0.24037
IVOL _s _maxadjF	0.34562	0.17472	3.80941	0.49905	0.23469
IVOL _s _CF	0.33505	0.36460	3.60390	0.50241	0.23519

Table 2. Cross-Sectional Forecast Accuracy Distribution for IVOL Models

This table presents the distributional performance of out-of-sample forecast errors for competing IVOL prediction models. Forecast errors are computed as the percentage increase in RMSE relative to the best-performing model in each month. Panel A reports results based on realized IVOL defined as the standard deviation of daily residuals from the Fama-French three-factor model, while Panel B uses the annualized square root of the sum of squared residuals. Reported values represent percentiles (10th, 25th, 50th, 75th, and 90th) of the cross-sectional distribution of relative forecast errors. Lower values indicate more accurate and consistent forecasts. Across both panels, models that exclude earnings announcement days (± 2 , ± 3 , ± 5) or extreme return days (MAX-adjusted) demonstrate improved accuracy over the baseline ARMA(1,1) specification (IVOL_F), with the ± 2 -day adjustment yielding the tightest error distribution.

Panel A: IVOL based on standard deviation of residuals

Model	10th	25th	50th	75th	90th
IVOL_F	0.004	0.012	0.033	0.085	0.193
IVOL_2F	0.003	0.009	0.026	0.066	0.156
IVOL_3F	0.004	0.011	0.027	0.068	0.154
IVOL_5F	0.008	0.021	0.047	0.097	0.190
IVOL_maxadjF	0.009	0.022	0.045	0.081	0.133

Panel B: IVOL based on the annualized square root of the sum of squared residuals

Model	10th	25th	50th	75th	90th
IVOL _s _F	0.003	0.011	0.031	0.081	0.185
IVOL _s _2F	0.003	0.009	0.024	0.062	0.147
IVOL _s _3F	0.004	0.010	0.025	0.063	0.144
IVOL _s _5F	0.008	0.020	0.046	0.094	0.184
IVOL _s _maxadjF	0.009	0.021	0.043	0.075	0.124

Table 3. Descriptive Statistics Across Realized IVOL Quintiles

This table reports average cross-sectional characteristics of stocks sorted into quintiles based on realized idiosyncratic volatility (IVOL), measured as the standard deviation of daily residuals from Fama-French regressions. In each month from January 1962 to December 2023, stocks are sorted into five quintiles based on their realized IVOL in month t . For each quintile, the median value of firm characteristics is computed within the month, and the final values reported are the time-series averages of these monthly medians. Reported variables include firm size (log market capitalization), beta, maximum daily return (MAX), momentum (MOM), return reversal (REV), and Amihud illiquidity (ILLIQ). The table also presents alternative IVOL measures excluding earnings announcement days ($\pm 2, \pm 3, \pm 5$), MAX-adjusted IVOL, and out-of-sample forecasts from six models: raw ARMA(1,1) (IVOL_F), EA-adjusted forecasts (IVOL_2F to IVOL_5F), MAX-adjusted forecast (maxadj_F), and a cross-sectional prediction (IVOL_CF). Each quintile contains on average 800 firms per month.

rank	IVOL	size	BETA	MAX	MOM	REV	ILLI	IVOL_2	IVOL_3
1	0.90	5.75	0.45	0.02	10.75	0.27	0.00	0.86	0.85
2	1.48	5.45	0.77	0.04	10.33	0.27	0.00	1.44	1.43
3	2.10	4.77	0.91	0.05	7.60	-0.05	0.01	2.05	2.05
4	3.01	4.07	1.00	0.08	1.21	-0.48	0.02	2.93	2.92
5	5.07	3.06	1.02	0.14	-14.90	0.32	0.06	4.92	4.89

Continue:

rank	IVOL	IVOL_5	maxadj	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	#share
1	0.90	0.01	0.01	1.21	1.16	1.15	1.15	1.05	800
2	1.48	0.01	0.01	1.63	1.55	1.54	1.54	1.41	800
3	2.10	0.02	0.02	2.12	2.03	2.02	2.02	1.83	800
4	3.01	0.03	0.03	2.78	2.67	2.65	2.65	2.37	800
5	5.07	0.05	0.04	3.99	3.88	3.85	3.83	3.37	800

Table 4. Portfolio Returns Sorted by Realized and Forecasted Idiosyncratic Volatility

This table reports average excess returns in month $t+1$ and corresponding risk-adjusted alphas for decile portfolios sorted by various measures of idiosyncratic volatility (IVOL), computed using the standard deviation of residuals from Fama-French regressions in month t . Each month from 1962 to 2023, stocks are sorted into deciles based on one of seven IVOL proxies: realized IVOL (IVOL_R), ARMA(1,1)-forecasted IVOL (IVOL_F), cross-sectional forecasted IVOL (IVOL_CF), EA-adjusted forecasts excluding ± 2 , ± 3 , and ± 5 days around earnings announcements (IVOL_2F, IVOL_3F, IVOL_5F), and MAX-adjusted IVOL forecasts (maxadj_F). Panel A presents equal-weighted portfolio returns, while Panel B shows value-weighted results. For each decile, we report the average return in month $t+1$ as well as alphas from the Fama-French six-factor model (α_{FF6}) and the q-factor model (α_{QF}), with Newey-West adjusted t-statistics in parentheses.

Panel A: Equal-Weighted							
	IVOL_R	IVOL_F	IVOL_CF	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F
Low	1.09	1.12	1.03	1.13	1.13	1.12	1.14
2	1.17	1.20	1.12	1.19	1.19	1.20	1.20
3	1.26	1.24	1.26	1.24	1.25	1.24	1.24
4	1.31	1.28	1.32	1.28	1.27	1.28	1.28
5	1.30	1.29	1.31	1.29	1.30	1.33	1.31
6	1.25	1.32	1.25	1.33	1.32	1.31	1.32
7	1.18	1.34	1.20	1.32	1.33	1.32	1.31
8	1.06	1.32	1.07	1.35	1.34	1.33	1.31
9	0.92	1.32	1.01	1.27	1.28	1.30	1.30
High	0.82	1.28	1.26	1.29	1.28	1.26	1.27
E[re]	-0.27	0.16	0.23	0.16	0.15	0.14	0.12
	-0.91	0.41	0.69	0.42	0.39	0.38	0.33
α_{FF6}	0.00	-0.19	-0.11	-0.19	-0.17	-0.14	-0.17
	-0.02	-1.51	-0.85	-1.47	-1.37	-1.10	-1.40
α_{QF}	-0.02	-0.22	-0.16	-0.23	-0.22	-0.19	-0.19
	-0.18	-1.65	-1.21	-1.72	-1.66	-1.40	-1.48
Panel B: Value-Weighted							
	IVOL_R	IVOL_F	IVOL_CF	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F
Low	0.97	0.95	0.95	0.96	0.95	0.94	0.97
2	0.98	1.04	1.01	1.02	1.03	1.00	1.03
3	1.01	1.06	1.05	1.06	1.07	1.06	0.99
4	1.00	1.02	1.10	1.02	1.00	1.06	1.09
5	1.00	1.14	0.95	1.10	1.18	1.14	1.13
6	1.06	1.09	0.89	1.11	1.05	1.16	1.15
7	0.83	1.12	0.74	1.08	1.07	1.02	0.99
8	0.63	0.76	0.53	0.77	0.81	0.82	0.79
9	0.41	0.68	0.15	0.68	0.66	0.70	0.65
High	-0.37	0.09	-0.27	0.09	0.13	0.17	0.14
E[re]	-1.34	-0.86	-1.22	-0.87	-0.82	-0.77	-0.84
	-3.92	-2.21	-3.95	-2.27	-2.18	-2.09	-2.16
α_{FF6}	-0.21	-0.16	-0.30	-0.22	-0.10	-0.14	-0.17
	-1.44	-0.95	-1.89	-1.19	-0.52	-0.70	-0.96
α_{QF}	-0.25	-0.21	-0.31	-0.27	-0.14	-0.25	-0.22
	-1.51	-1.19	-1.87	-1.27	-0.68	-1.11	-1.04

Table 5. Portfolio Returns Sorted by Realized and Forecasted IVOL (RSS-based Measure)

This table presents portfolio performance for deciles sorted by various idiosyncratic volatility (IVOL) measures computed as the annualized square root of the sum of squared residuals (RSS) from Fama-French regressions. Each month from 1962 to 2023, stocks are sorted into deciles based on one of the seven IVOL measures in month t , and average excess returns in month $t+1$ are computed. The columns include realized IVOL (IVOLs_R), ARMA(1,1) forecast (IVOLs_F), cross-sectional forecast (IVOLs_CF), EA-adjusted forecasts excluding ± 2 , ± 3 , and ± 5 days (IVOLs_2F, IVOLs_3F, IVOLs_5F), and the MAX-adjusted forecast (s_maxadjF). Panel A reports equal-weighted returns, while Panel B shows value-weighted returns. We also report alphas from the Fama-French six-factor model (α_{FF6}) and q-factor model (α_{QF}), with Newey-West adjusted t -statistics in parentheses.

Panel A: Equal-Weighted

	IVOLs_R	IVOLs_F	IVOLs_CF	IVOLs_2F	IVOLs_3F	IVOLs_5F	s_maxadjF
Low	1.09	1.12	1.03	1.13	1.13	1.12	1.14
2	1.17	1.18	1.12	1.18	1.18	1.19	1.19
3	1.26	1.25	1.25	1.26	1.26	1.25	1.25
4	1.31	1.27	1.32	1.27	1.26	1.28	1.27
5	1.30	1.30	1.31	1.31	1.33	1.35	1.33
6	1.25	1.33	1.26	1.33	1.31	1.31	1.32
7	1.18	1.33	1.20	1.32	1.32	1.32	1.29
8	1.06	1.32	1.07	1.36	1.35	1.34	1.30
9	0.92	1.31	1.01	1.26	1.27	1.29	1.30
High	0.82	1.28	1.26	1.29	1.29	1.25	1.27
E[re]	-0.27	0.16	0.23	0.17	0.16	0.13	0.13
	-0.91	0.43	0.69	0.44	0.42	0.35	0.35
α_{FF6}	0.00	-0.18	-0.11	-0.17	-0.14	-0.12	-0.16
	-0.02	-1.47	-0.85	-1.29	-1.12	-0.94	-1.29
α_{QF}	-0.02	-0.21	-0.16	-0.21	-0.19	-0.17	-0.17
	-0.18	-1.60	-1.21	-1.56	-1.44	-1.29	-1.32

Panel B: Value-Weighted

	IVOLs_R	IVOLs_F	IVOLs_CF	IVOLs_2F	IVOLs_3F	IVOLs_5F	s_maxadjF
Low	0.97	0.96	0.95	0.96	0.95	0.95	0.96
2	0.98	1.01	1.00	1.01	1.03	0.97	1.04
3	1.01	1.04	1.05	1.07	1.07	1.05	1.00
4	1.00	1.03	1.10	1.03	1.03	1.11	1.06
5	1.00	1.14	0.95	1.11	1.18	1.12	1.16
6	1.06	1.09	0.89	1.11	1.04	1.14	1.11
7	0.83	1.11	0.74	1.06	1.06	1.01	1.01
8	0.63	0.78	0.53	0.80	0.87	0.84	0.75
9	0.41	0.63	0.14	0.63	0.63	0.67	0.68
High	-0.37	0.06	-0.27	0.09	0.09	0.15	0.14
E[re]	-1.34	-0.90	-1.22	-0.87	-0.86	-0.79	-0.82
	-3.92	-2.31	-3.95	-2.24	-2.26	-2.17	-2.10
α_{FF6}	-0.21	-0.15	-0.30	-0.19	-0.08	-0.14	-0.14
	-1.44	-0.84	-1.89	-1.01	-0.40	-0.68	-0.76
α_{QF}	-0.25	-0.21	-0.31	-0.25	-0.14	-0.26	-0.19
	-1.51	-1.09	-1.86	-1.21	-0.66	-1.16	-0.94

Table 6. Portfolio Returns for Event-Driven vs. Non-Event-Driven Volatility

This table compares portfolio returns across volatility measures that isolate event-driven and non-event-driven components of realized IVOL. Each month from 1962 to 2023, stocks are sorted into deciles based on IVOL variants: IVOL within ± 2 , ± 3 , and ± 5 days around earnings announcements (EAD_2, EAD_3, EAD_5), and maximum-day Residual (MAX Day); as well as IVOL calculated excluding those days (IVOL_2, IVOL_3, IVOL_5, Maxadj). Panel A presents equal-weighted average returns and alphas; Panel B shows value-weighted results.

Panel A: Equal-Weighted

	Earning Announcement Window or MAX Day				Out of Earning Announcement or MAX Window			
	EAD_2	EAD_3	EAD_5	MAX Day	IVOL_2	IVOL_3	IVOL_5	Maxadj
Low	1.18	1.19	1.16	1.12	1.10	1.09	1.09	1.12
2	1.12	1.10	1.01	1.20	1.17	1.18	1.19	1.17
3	1.05	1.04	1.04	1.23	1.27	1.26	1.25	1.22
4	1.09	1.07	1.12	1.29	1.31	1.32	1.32	1.28
5	1.17	1.18	1.19	1.30	1.31	1.32	1.30	1.28
6	1.13	1.12	1.17	1.27	1.27	1.27	1.24	1.23
7	1.20	1.22	1.25	1.16	1.17	1.17	1.17	1.15
8	1.22	1.26	1.30	1.10	1.07	1.07	1.07	1.05
9	1.17	1.18	1.17	1.01	0.90	0.90	0.91	0.91
High	1.02	0.99	0.96	0.72	0.78	0.79	0.86	0.90
E[re]	-0.17	-0.20	-0.20	-0.40	-0.32	-0.31	-0.23	-0.22
	-1.13	-1.27	-1.27	-1.61	-1.06	-1.02	-0.76	-0.74
α_{FF6}	-0.41	-0.40	-0.40	-0.08	-0.01	0.00	0.00	-0.04
	-4.48	-4.10	-3.87	-0.89	-0.07	0.02	-0.01	-0.29
α_{QF}	-0.40	-0.39	-0.36	-0.11	-0.02	0.00	-0.02	-0.05
	-4.19	-4.02	-3.04	-1.21	-0.13	0.01	-0.15	-0.34

Panel B: Value-Weighted

	Earning Announcement Window or MAX Day				Out of Earning Announcement Window or MAX Day			
	EAD_2	EAD_3	EAD_5	MAX Day	IVOL_2	IVOL_3	IVOL_5	Maxadj
Low	0.96	0.95	0.93	0.89	0.98	0.97	0.99	0.98
2	0.86	0.90	0.92	1.02	0.97	0.99	1.03	0.97
3	1.04	0.99	0.98	0.93	1.03	1.03	0.98	1.02
4	0.96	0.82	0.85	0.99	1.02	1.02	1.12	0.99
5	0.86	0.94	0.86	1.03	1.04	1.04	1.02	1.03
6	0.99	1.00	1.06	1.03	1.06	1.05	0.94	1.05
7	0.96	0.99	0.94	0.90	0.77	0.77	0.79	0.83
8	0.77	0.88	0.95	0.92	0.66	0.65	0.61	0.64
9	1.04	1.05	1.02	0.60	0.34	0.37	0.18	0.34
High	0.91	0.78	0.77	0.09	-0.43	-0.43	-0.41	-0.34
E[re]	-0.05	-0.17	-0.16	-0.79	-1.40	-1.39	-1.40	-1.32
	-0.39	-1.02	-0.88	-3.02	-4.21	-4.18	-4.35	-3.95
α_{FF6}	-0.46	-0.42	-0.43	-0.17	-0.17	-0.19	-0.16	-0.14
	-3.18	-2.77	-2.75	-1.55	-1.04	-1.25	-0.99	-0.86
α_{QF}	-0.36	-0.33	-0.34	-0.16	-0.21	-0.22	-0.17	-0.15
	-2.73	-2.45	-2.39	-1.20	-1.15	-1.30	-1.03	-0.88

Table 7. Frequency of Forecast Overestimation vs. Underestimation Across IVOL Models

This table reports the proportion of monthly firm-level observations for which forecasted idiosyncratic volatility (IVOL) exceeds or falls below realized IVOL, based on different forecasting specifications. For each model—ARMA(1,1) applied to raw residuals (IVOL_F), event-adjusted residuals (IVOL_2F, IVOL_3F, IVOL_5F), and MAX-adjusted residuals (Maxadj_F), the table reports the percentage of observations where the forecast overestimates (Realized < Forecast) or underestimates (Realized > Forecast) realized volatility.

	Realized < Forecast	Realized > Forecast
IVOL_F	60%	40%
IVOL_2F	56%	44%
IVOL_3F	56%	44%
IVOL_5F	56%	44%
Maxadj_F	46%	54%

Table 8. Portfolio Returns Following IVOL Forecast Overestimation (Forecast IVOL > Realized IVOL)

This table presents the performance of decile-sorted portfolios in months where forecasted idiosyncratic volatility (IVOL) exceeds realized IVOL, i.e., forecast overestimation. Stocks are sorted into deciles based on forecasted IVOL levels, and average monthly returns are reported for both equal-weighted (Panel A) and value-weighted (Panel B) portfolios from month $t+1$. Columns correspond to six forecasting models: ARMA(1,1) on raw residuals (IVOL_F), event-adjusted forecasts (IVOL_2F, IVOL_3F, IVOL_5F), forecasts with maximum-day residuals removed (Maxadj_F), and a cross-sectional model (IVOL_CF). For each column, the table reports the average raw excess return ($E[re]$), the alpha with respect to the Fama-French six-factor model (α_{FF6}), and the alpha with respect to the Q-factor model (α_{QF}), with Newey-West t-statistics in parentheses.

Panel A: Equal-Weighted

	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Low	0.71	0.71	0.70	0.70	0.61	0.81
2	0.70	0.68	0.68	0.67	0.60	0.71
3	0.62	0.60	0.61	0.60	0.51	0.65
4	0.61	0.57	0.55	0.56	0.44	0.53
5	0.44	0.40	0.42	0.38	0.25	0.34
6	0.28	0.19	0.18	0.15	0.10	0.12
7	-0.06	-0.14	-0.16	-0.13	-0.30	-0.23
8	-0.51	-0.63	-0.65	-0.65	-0.86	-0.79
9	-1.16	-1.29	-1.32	-1.27	-1.50	-1.50
High	-1.94	-2.10	-2.12	-1.95	-2.54	-2.42
$E[re]$	-2.65	-2.81	-2.82	-2.65	-3.14	-3.23
	-8.00	-8.16	-8.25	-8.04	-10.13	-11.53
α_{FF6}	-0.58	-0.64	-0.62	-0.67	-0.69	-0.89
	-5.35	-6.40	-6.15	-6.43	-6.12	-8.42
α_{QF}	-0.61	-0.68	-0.66	-0.70	-0.72	-0.92
	-5.24	-5.92	-5.78	-5.95	-5.40	-7.90

Panel B: Value-Weighted

	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Low	0.97	0.97	0.96	0.91	0.91	0.81
2	0.92	0.85	0.89	0.90	0.90	0.76
3	0.89	0.91	0.91	0.92	0.85	0.74
4	1.02	1.04	1.01	1.12	0.88	0.70
5	0.89	0.91	0.90	0.90	0.84	0.53
6	0.83	0.74	0.74	0.81	0.77	0.36
7	0.37	0.34	0.33	0.30	0.38	0.01
8	-0.01	-0.18	-0.18	-0.02	-0.32	-0.55
9	-0.58	-0.72	-0.80	-0.55	-0.85	-1.07
High	-1.36	-1.49	-1.45	-1.27	-1.71	-1.90
$E[re]$	-2.33	-2.47	-2.41	-2.18	-2.62	-2.70
	-7.44	-7.68	-7.52	-7.63	-8.73	-11.77
α_{FF6}	-0.37	-0.35	-0.33	-0.31	-0.33	-0.65
	-2.08	-2.12	-1.99	-1.74	-1.83	-4.83
α_{QF}	-0.46	-0.40	-0.40	-0.39	-0.40	-0.64
	-2.47	-2.06	-2.18	-1.96	-2.04	-4.49

Table 9. Portfolio Returns Following IVOL Forecast Underestimation (Forecast IVOL < Realized IVOL)

This table reports portfolio performance in months where forecasted idiosyncratic volatility (IVOL) is lower than realized IVOL, reflecting underestimation of volatility. Stocks are sorted into deciles based on the forecasted IVOL measure at time t , and equal-weighted (Panel A) and value-weighted (Panel B) portfolio returns are reported for the next month. Forecasting methods include ARMA(1,1) applied to raw IVOLs (IVOL_F), earnings-adjusted models (IVOL_2F, IVOL_3F, IVOL_5F), the MAX-adjusted specification (Maxadj_F), and a cross-sectional prediction model (IVOL_CF). Each panel displays average excess returns (E[re]), Fama-French six-factor alphas (α_{FF6}), and Q-factor model alphas (α_{QF}), with Newey-West adjusted t-statistics in parentheses.

Panel A: Equal-Weighted

	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Low	1.61	1.58	1.56	1.53	1.49	1.16
2	1.78	1.74	1.72	1.71	1.61	1.60
3	1.99	1.93	1.94	1.92	1.75	1.99
4	2.21	2.12	2.10	2.12	1.89	2.29
5	2.45	2.37	2.32	2.32	2.11	2.47
6	2.76	2.56	2.55	2.54	2.18	2.61
7	3.08	2.95	2.93	2.86	2.43	2.86
8	3.73	3.49	3.46	3.38	2.85	3.07
9	4.63	4.35	4.29	4.22	3.28	3.82
High	6.76	6.32	6.20	6.10	4.50	5.58
E[re]	5.15	4.74	4.64	4.57	3.01	4.42
	7.17	7.23	7.21	7.13	5.36	7.42
α_{FF6}	0.48	0.44	0.41	0.43	0.29	0.94
	2.75	2.62	2.43	2.51	1.86	4.24
α_{QF}	0.44	0.39	0.37	0.36	0.30	0.84
	2.45	2.21	2.11	1.99	1.77	3.71

Panel B: Value-Weighted

	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Low	0.98	1.01	1.03	1.02	1.00	0.98
2	1.10	1.15	1.07	1.14	1.11	1.23
3	1.18	1.18	1.20	1.24	1.09	1.39
4	1.27	1.22	1.26	1.20	1.25	1.63
5	1.55	1.46	1.53	1.52	1.31	1.63
6	1.63	1.62	1.56	1.70	1.43	1.65
7	2.15	2.03	2.04	1.87	1.70	1.78
8	2.48	2.37	2.26	2.11	1.88	1.85
9	2.94	2.89	2.99	2.93	2.10	2.06
High	4.20	3.81	3.71	3.54	2.56	2.97
E[re]	3.22	2.80	2.68	2.52	1.57	1.99
	4.39	4.19	4.08	3.70	2.57	3.08
α_{FF6}	0.14	0.03	0.12	0.07	-0.03	0.31
	0.49	0.11	0.47	0.25	-0.13	1.31
α_{QF}	0.14	-0.04	0.06	-0.01	-0.06	0.27
	0.45	-0.15	0.23	-0.02	-0.24	1.04

Table 10. Fama–MacBeth Regressions of Returns on Idiosyncratic Volatility Measures

This table reports results from monthly **univariate** Fama–MacBeth (1973) regressions of one-month-ahead stock returns on alternative measures of idiosyncratic volatility. Panel A reports results using IVOL computed as the standard deviation of residuals from the Fama–French six-factor model. Panel B uses the annualized square root of the sum of squared residuals as the IVOL definition. Panel C decomposes realized IVOL into event-driven (EAD ± 2 , ± 3 , ± 5 , and MAX day) and non-event-driven components (NO-EAD and non-MAX windows). Coefficients and corresponding Newey–West (1987) t-statistics (in italics) are reported.

Panel A: IVOL based on standard deviation of residuals

	IVOL_R	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Intercept	1.39	1.21	1.21	1.20	1.21	1.16	1.22
	7.77	7.45	7.29	7.23	7.37	6.68	7.35
Coeff	-11.59	-2.80	-3.05	-2.80	-2.94	-1.75	-5.54
	-2.12	-0.32	-0.34	-0.31	-0.34	-0.15	-0.63

Panel B: IVOL based on the annualized square root of the sum of squared residuals

	IVOLs_R	IVOLs_F	IVOLs_2F	IVOLs_3F	IVOLs_5F	s_Maxadj_F	IVOLs_CF
Intercept	1.36	1.22	1.20	1.20	1.22	1.17	1.22
	7.77	7.45	7.28	7.25	7.41	6.73	7.35
Coeff	-0.75	-0.20	-0.19	-0.20	-0.22	-0.15	-0.36
	-2.12	-0.34	-0.33	-0.33	-0.39	-0.21	-0.63

Panel C: Realized IVOL around Earning Announcement window and out of EA window

	EAD_2	EAD_3	EAD_5	MAX DAY	IVOL_2	IVOL_3	IVOL_5	Maxadj
Intercept	1.16	1.16	1.16	1.34	1.37	1.37	1.37	1.31
	5.14	5.13	5.18	6.82	7.83	7.82	7.77	7.70
Coeff	-0.28	1.18	1.00	-4.89	-12.57	-12.64	-12.51	-11.98
	-0.14	0.57	0.46	-3.14	-2.26	-2.28	-2.27	-1.64

Table 11. Double-Sorted Portfolios Based on Maximum Daily Return (MAX) and Realized Idiosyncratic Volatility (IVOL)

This table reports value-weighted and equal-weighted average monthly returns and risk-adjusted alphas for 25 portfolios double sorted on MAX and realized IVOL. Each month from July 1962 to December 2023, stocks are sorted into quintiles based on MAX (maximum daily return over the prior month), and within each MAX quintile, stocks are further sorted into quintiles based on realized IVOL (standard deviation of residuals from the Fama–French three-factor model). Panel A presents results for equal-weighted portfolios; Panel B presents value-weighted results. For each cell, the table reports average monthly return, followed by the Fama–French five-factor plus momentum alpha and the Q-factor alpha in subsequent columns. Newey–West (1987) adjusted t-statistics are shown in parentheses. The bottom row and rightmost column report return and alpha spreads between high and low IVOL and MAX quintiles, respectively.

	Low-IVOL	2	3	4	High-IVOL	E[re]	α_{FF6}	α_{QF}
PANEL (A) : Equal-Weighted (1962-2023)								
Low-MAX	1.01	1.03	1.10	1.23	1.70	0.69 (5.17)	0.41 (3.02)	0.33 (2.22)
2	1.16	1.20	1.32	1.48	1.61	0.45 (2.82)	0.08 (0.53)	0.03 (0.20)
3	1.14	1.19	1.24	1.26	1.50	0.36 (1.84)	0.03 (0.14)	-0.07 (-0.34)
4	1.15	1.05	0.97	1.02	1.37	0.22 (1.03)	-0.19 (-0.87)	-0.20 (-0.94)
High-MAX	0.83	0.65	0.57	0.60	0.72	-0.11 (-0.39)	-0.65 (-2.23)	-0.60 (-1.82)
E[re]	-0.18 (-0.82)	-0.38 (-1.50)	-0.54 (-1.82)	-0.63 (-2.10)	-0.98 (-2.94)			
α_{FF6}	-0.37 (-1.68)	-0.65 (-2.56)	-0.85 (-2.66)	-1.00 (-3.06)	-1.43 (-4.25)			
α_{QF}	-0.45 (-1.90)	-0.69 (-2.75)	-0.93 (-3.15)	-1.05 (-3.53)	-1.38 (-3.79)			
PANEL (B): Value-Weighted (1962-2023)								
Low-MAX	0.82	0.94	0.99	1.08	1.04	0.22 (1.73)	-0.05 (-0.35)	-0.05 (-0.37)
2	0.96	0.92	1.04	1.08	0.99	0.03 (0.22)	-0.34 (-1.89)	-0.36 (-1.80)
3	0.92	1.01	1.09	0.93	0.82	-0.10 (-0.57)	-0.41 (-1.97)	-0.49 (-2.00)
4	1.12	0.85	0.65	0.60	0.33	-0.79 (-3.47)	-1.14 (-4.57)	-1.10 (-4.38)
High-MAX	0.64	0.44	0.17	-0.21	-1.12	-1.76 (-5.09)	-2.15 (-5.53)	-1.97 (-5.39)
E[re]	-0.18 (-0.77)	-0.50 (-1.51)	-0.82 (-2.32)	-1.29 (-3.85)	-2.16 (-5.42)			
α_{FF6}	-0.35 (-1.29)	-0.70 (-2.17)	-1.04 (-2.46)	-1.61 (-4.25)	-2.45 (-5.09)			
α_{QF}	-0.48 (-1.63)	-0.77 (-2.23)	-1.24 (-3.28)	-1.64 (-4.18)	-2.40 (-5.37)			

Table 12. Double-Sorted Portfolios Based on Maximum Daily Return (MAX) and Forecasted Idiosyncratic Volatility (IVOL)

This table presents average monthly returns and risk-adjusted alphas for portfolios double sorted on MAX and forecasted IVOL. Each month from July 1962 to December 2023, stocks are first sorted into quintiles based on MAX (the highest daily return in the previous month), and then into quintiles based on IVOL forecasts from ARMA(1,1) models applied to residuals (IVOL_F). Panel A shows equal-weighted portfolio results, while Panel B shows value-weighted results. For each cell, the average monthly return is reported, followed by the Fama–French Five-factor plus momentum alpha (α_{FF6}) and Q-factor alpha (α_{QF}). Newey–West (1987) adjusted t-statistics are reported in parentheses.

	Low-IVOL_F	2	3	4	High-IVOL_F	E[re]	α_{FF6}	α_{QF}
PANEL (A) : Equal-Weighted (1962-2023)								
Low-MAX	1.02	1.15	1.14	1.32	1.73	0.71 (3.78)	0.27 (1.62)	0.27 (1.68)
2	1.12	1.23	1.33	1.45	1.69	0.57 (2.75)	0.14 (0.75)	0.09 (0.49)
3	1.14	1.23	1.31	1.45	1.68	0.54 (2.26)	0.12 (0.52)	0.04 (0.17)
4	1.13	1.16	1.21	1.38	1.56	0.43 (1.42)	-0.04 (-0.14)	-0.15 (-0.56)
High-MAX	0.93	0.94	0.90	0.75	1.25	0.33 (0.79)	-0.29 (-0.74)	-0.26 (-0.60)
E[re]	-0.09 (-0.58)	-0.21 (-0.87)	-0.24 (-0.89)	-0.57 (-1.82)	-0.47 (-1.25)			
α_{FF6}	-0.41 (-2.23)	-0.57 (-2.14)	-0.62 (-2.10)	-0.93 (-2.98)	-0.97 (-2.60)			
α_{QF}	-0.46 (-2.51)	-0.68 (-2.70)	-0.71 (-2.52)	-0.95 (-3.09)	-0.99 (-2.43)			
PANEL (B): Value-Weighted (1962-2023)								
Low-MAX	0.91	1.04	0.97	1.24	1.25	0.34 (2.08)	0.01 (0.09)	-0.03 (-0.15)
2	0.92	1.00	1.09	1.21	1.19	0.27 (1.27)	-0.14 (-0.65)	-0.16 (-0.75)
3	1.04	0.95	1.12	1.23	1.26	0.22 (0.83)	-0.04 (-0.13)	-0.21 (-0.75)
4	0.94	1.02	1.01	0.65	0.54	-0.40 (-1.46)	-0.86 (-3.10)	-1.05 (-3.67)
High-MAX	0.84	0.57	0.24	0.16	-0.73	-1.57 (-3.35)	-2.17 (-4.71)	-1.94 (-4.19)
E[re]	-0.07 (-0.29)	-0.47 (-1.65)	-0.74 (-2.22)	-1.08 (-2.93)	-1.98 (-4.30)			
α_{FF6}	-0.37 (-1.34)	-0.76 (-2.31)	-0.99 (-2.61)	-1.53 (-3.57)	-2.56 (-5.67)			
α_{QF}	-0.59 (-1.88)	-0.93 (-2.71)	-1.13 (-2.84)	-1.79 (-4.34)	-2.50 (-5.94)			

Table 13

Double sorted portfolios based on maximum daily return and Forecasted IVOL_EA2

	Low-IVOL_F	2	3	4	High-IVOL_F	E[re]	α_{FF6}	α_{QF}
PANEL (A) : Equal-Weighted (1962-2023)								
Low-MAX	1.02	1.15	1.16	1.30	1.72	0.70 (3.76)	0.26 (1.53)	0.23 (1.42)
2	1.12	1.19	1.35	1.46	1.69	0.56 (2.74)	0.12 (0.66)	0.07 (0.36)
3	1.15	1.25	1.30	1.47	1.65	0.51 (2.12)	0.10 (0.41)	0.01 (0.02)
4	1.12	1.16	1.25	1.38	1.51	0.39 (1.29)	-0.08 (-0.27)	-0.20 (-0.74)
High-MAX	0.94	0.93	0.86	0.84	1.21	0.28 (0.67)	-0.35 (-0.91)	-0.32 (-0.77)
E[re]	-0.09 (-0.55)	-0.22 (-0.91)	-0.30 (-1.12)	-0.46 (-1.46)	-0.51 (-1.37)			
α_{FF6}	-0.41 (-2.27)	-0.59 (-2.29)	-0.67 (-2.30)	-0.82 (-2.51)	-1.01 (-2.75)			
α_{QF}	-0.46 (-2.48)	-0.65 (-2.56)	-0.80 (-2.98)	-0.86 (-2.72)	-1.00 (-2.54)			
PANEL (B): Value-Weighted (1962-2023)								
Low-MAX	0.91	1.05	0.96	1.21	1.20	0.29 (1.83)	-0.13 (-0.76)	-0.20 (-1.14)
2	0.90	1.04	1.05	1.20	1.32	0.42 (1.95)	-0.01 (-0.04)	0.04 (0.15)
3	1.02	1.03	1.07	1.22	1.17	0.16 (0.60)	-0.12 (-0.40)	-0.26 (-0.93)
4	0.93	1.08	0.96	0.62	0.48	-0.45 (-1.51)	-0.80 (-2.55)	-1.05 (-3.15)
High-MAX	0.83	0.53	0.34	0.21	-0.88	-1.71 (-3.98)	-2.35 (-5.71)	-2.14 (-5.28)
E[re]	-0.09 (-0.38)	-0.52 (-1.72)	-0.62 (-1.75)	-1.00 (-2.67)	-2.08 (-4.98)			
α_{FF6}	-0.40 (-1.50)	-0.78 (-2.33)	-0.89 (-2.13)	-1.46 (-3.34)	-2.62 (-6.54)			
α_{QF}	-0.60 (-1.93)	-1.02 (-3.08)	-1.11 (-2.62)	-1.52 (-3.63)	-2.54 (-6.57)			

Table 14

Double sorted portfolios based on maximum daily return and Realized IVOL-NOEAD-5

	LOW-NOEAD_5	2	3	4	High- NOEAD_5	E[re]	α_{FF6}	α_{QF}
PANEL (A) : Equal-Weighted (1962-2023)								
Low-MAX	1.01	1.03	1.08	1.19	1.73	0.72 (5.30)	0.43 (3.12)	0.36 (2.34)
2	1.17	1.20	1.32	1.48	1.63	0.46 (2.89)	0.08 (0.54)	0.02 (0.11)
3	1.15	1.19	1.21	1.29	1.55	0.40 (2.15)	0.01 (0.06)	-0.09 (-0.47)
4	1.17	1.01	0.98	1.02	1.38	0.21 (1.01)	-0.24 (-1.11)	-0.26 (-1.22)
High-MAX	0.89	0.60	0.62	0.53	0.79	-0.10 (-0.34)	-0.72 (-2.30)	-0.68 (-1.95)
E[re]	-0.12 (-0.53)	-0.42 (-1.73)	-0.46 (-1.55)	-0.65 (-2.23)	-0.94 (-2.76)			
α_{FF6}	-0.26 (-1.10)	-0.68 (-2.80)	-0.80 (-2.52)	-1.02 (-3.28)	-1.42 (-3.99)			
α_{QF}	-0.34 (-1.38)	-0.74 (-2.97)	-0.84 (-2.80)	-1.07 (-3.70)	-1.38 (-3.67)			
PANEL (B): Value-Weighted (1962-2023)								
Low-MAX	0.80	0.91	1.05	1.14	1.23	0.42 (3.29)	0.18 (1.18)	0.21 (1.27)
2	0.98	0.89	1.19	1.15	1.06	0.08 (0.43)	-0.31 (-1.53)	-0.34 (-1.53)
3	0.92	1.00	0.99	1.13	0.81	-0.10 (-0.49)	-0.51 (-2.27)	-0.62 (-2.50)
4	1.06	0.71	0.69	0.45	0.15	-0.90 (-4.33)	-1.31 (-5.41)	-1.28 (-5.26)
High-MAX	0.61	0.22	0.02	-0.47	-1.17	-1.78 (-5.88)	-2.26 (-6.27)	-2.31 (-6.60)
E[re]	-0.20 (-0.84)	-0.70 (-2.27)	-1.03 (-3.13)	-1.61 (-4.50)	-2.40 (-6.36)			
α_{FF6}	-0.30 (-1.22)	-0.79 (-2.36)	-1.22 (-3.20)	-1.98 (-4.62)	-2.74 (-5.94)			
α_{QF}	-0.36 (-1.36)	-1.04 (-3.03)	-1.37 (-3.77)	-1.98 (-4.71)	-2.88 (-6.30)			

Table 15

Double sorted portfolios based on maximum daily return and MAX-IVOL RATIO

	Low_ratio	2	3	4	High_ratio	E[re]	α_{FF6}	α_{QF}
PANEL (A) : Equal-Weighted (1962-2023)								
Low_first	1.21	1.22	1.25	1.23	1.16	-0.05 (-0.50)	-0.41 (-3.87)	-0.38 (-3.51)
2	1.34	1.39	1.37	1.33	1.33	-0.02 (-0.23)	-0.45 (-4.80)	-0.38 (-3.64)
3	1.25	1.28	1.29	1.29	1.23	-0.02 (-0.13)	-0.45 (-3.35)	-0.32 (-2.58)
4	1.06	1.08	1.05	1.19	1.18	0.11 (0.72)	-0.30 (-1.93)	-0.25 (-1.43)
High_first	0.81	0.73	0.66	0.62	0.56	-0.25 (-1.45)	-0.62 (-3.17)	-0.43 (-2.09)
E[re]	-0.40 (-1.42)	-0.49 (-1.80)	-0.59 (-2.15)	-0.61 (-2.26)	-0.61 (-2.51)			
α_{FF6}	-0.72 (-2.27)	-0.84 (-2.97)	-0.91 (-3.27)	-0.89 (-3.06)	-0.93 (-3.85)			
α_{QF}	-0.88 (-2.90)	-0.88 (-3.18)	-0.94 (-3.40)	-0.89 (-3.15)	-0.93 (-3.63)			
PANEL (B) : Equal-Weighted (1962-2023)								
Low_first	0.80	1.03	0.98	0.98	1.01	0.20 (2.26)	-0.08 (-0.98)	-0.25 (-2.97)
2	0.86	0.96	1.04	1.04	1.00	0.14 (1.57)	-0.17 (-1.65)	-0.11 (-0.91)
3	0.94	0.94	1.00	0.99	1.02	0.09 (0.55)	-0.39 (-2.49)	-0.34 (-1.91)
4	0.59	0.91	0.75	1.05	1.02	0.43 (2.24)	-0.10 (-0.50)	-0.10 (-0.45)
High_first	0.10	0.02	0.00	0.46	0.54	0.44 (1.42)	-0.12 (-0.39)	0.09 (0.31)
E[re]	-0.70 (-1.86)	-1.01 (-2.75)	-0.99 (-3.50)	-0.52 (-1.99)	-0.47 (-2.08)			
α_{FF6}	-0.85 (-2.20)	-1.20 (-3.16)	-1.20 (-3.98)	-0.73 (-2.62)	-0.89 (-3.27)			
α_{QF}	-1.16 (-3.00)	-1.29 (-3.65)	-1.44 (-4.49)	-0.71 (-2.42)	-0.82 (-2.84)			

Table 16. Average Monthly Correlation Matrices: Full Sample

This table reports the time-series average of monthly cross-sectional Pearson correlation matrices computed from January 1962 to December 2023. Panel A includes correlations among realized IVOL variants (baseline, earnings-adjusted, and max-adjusted) and event-based volatility measures. Panel B includes forecasted IVOLs derived from ARMA(1,1), event- and max-adjusted models, and a cross-sectional predictive model, as well as their relationship with realized IVOL and the MAX. Panel C presents correlations between realized IVOL and firm characteristics including beta, size, momentum, skewness, illiquidity, and the Resid square of MAX day divided by IVOL MAXADJ squared (ratio). Correlations are computed within each month using pairwise deletion and then averaged across months.

Panel A: Correlation matrix between different IVOL metrics

	IVOL	IVOL_2	IVOL_3	IVOL_5	EAD_2	EAD_3	EAD_5	Maxadj	EAD_max	Lead_IVOL	MAX
IVOL	1										
IVOL_2	0.98	1									
IVOL_3	0.98	0.99	1								
IVOL_5	0.98	0.99	0.99	1							
EAD_2	0.77	0.49	0.47	0.42	1						
EAD_3	0.8	0.57	0.51	0.45	0.94	1					
EAD_5	0.87	0.74	0.7	0.5	0.84	0.89	1				
Maxadj	0.91	0.91	0.91	0.91	0.71	0.74	0.8	1			
EAD_max	0.89	0.88	0.87	0.88	0.67	0.7	0.76	0.67	1		
Lead_IVOL	0.57	0.57	0.57	0.57	0.42	0.45	0.51	0.61	0.44	1	
MAX	0.89	0.87	0.87	0.87	0.67	0.7	0.76	0.68	0.94	0.43	1

Panel B: Correlation matrix between IVOL and IVOL forecasts

	IVOL	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	Cross_F	Lead_IVOL	MAX
IVOL	1								
IVOL_F	0.66	1							
IVOL_2F	0.66	0.99	1						
IVOL_3F	0.66	0.98	0.99	1					
IVOL_5F	0.65	0.95	0.96	0.96	1				
Maxadj_F	0.68	0.95	0.94	0.94	0.92	1			
Cross_F	0.91	0.72	0.72	0.72	0.71	0.74	1		
Lead_IVOL	0.57	0.64	0.64	0.64	0.62	0.65	0.62	1	
MAX	0.89	0.51	0.51	0.51	0.5	0.52	0.69	0.43	1

Panel C: Correlation matrix between IVOL and other factors

	IVOL	BETA	Size	Reversal	MOM	ILLI	ISKEW	Ratio	Lead_IVOL	MAX
IVOL	1									
BETA	0.07	1								
Size	-0.42	0.08	1							
Reversal	0.16	0.02	-0.01	1						
MOM	-0.12	0.02	0.17	0.01	1					
ILLI	0.33	-0.02	-0.28	-0.01	-0.08	1				
ISKEW	0.18	0.02	-0.19	0.11	0.19	0.06	1			
Ratio	0.21	-0.02	-0.09	0.17	-0.02	0.02	0.16	1		
Lead_IVOL	0.57	0.04	-0.41	-0.05	-0.12	0.32	0.11	0.02	1	
MAX	0.89	0.11	-0.32	0.35	-0.1	0.24	0.23	0.34	0.43	1

Table 17. Average Monthly Correlation Matrices: IVOL Forecast Overestimation

This table reports average monthly cross-sectional correlation matrices conditional on the forecasted IVOL (IVOL_F) exceeding the realized IVOL (Lead_IVOL) in month t+1 (i.e., overestimation). The sample is restricted to firm-months where $IVOL_forecast > Lead_IVOL$. Panel A shows the correlation structure among realized IVOL measures and event-driven components. Panel B focuses on the internal relationships among forecasted IVOLs and their alignment with realized IVOL and MAX. Panel C displays correlations between realized IVOL and firm characteristics under overestimation. All correlations are computed monthly and averaged across time.

Panel A: Correlation matrix between different IVOL metrics

	IVOL	IVOL_2	IVOL_3	IVOL_5	EAD_2	EAD_3	EAD_5	Maxadj	EAD_max	Lead_IVOL	MAX
IVOL	1										
IVOL_2	0.97	1									
IVOL_3	0.97	0.99	1								
IVOL_5	0.97	0.99	0.99	1							
EAD_2	0.77	0.48	0.46	0.42	1						
EAD_3	0.81	0.57	0.5	0.45	0.94	1					
EAD_5	0.87	0.73	0.7	0.5	0.84	0.89	1				
Maxadj	0.91	0.9	0.89	0.89	0.7	0.74	0.8	1			
EAD_max	0.89	0.86	0.86	0.87	0.68	0.71	0.76	0.66	1		
Lead_IVOL	0.67	0.67	0.67	0.66	0.49	0.53	0.58	0.73	0.5	1	
MAX	0.89	0.86	0.86	0.86	0.67	0.7	0.76	0.67	0.94	0.51	1

Panel B: Correlation matrix between IVOL and IVOL forecasts

	IVOL	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	Cross_F	Lead_IVOL	MAX
IVOL	1								
IVOL_F	0.68	1							
IVOL_2F	0.68	0.99	1						
IVOL_3F	0.68	0.98	0.99	1					
IVOL_5F	0.66	0.95	0.96	0.96	1				
Maxadj_F	0.71	0.94	0.94	0.94	0.91	1			
Cross_F	0.91	0.73	0.73	0.73	0.71	0.75	1		
Lead_IVOL	0.67	0.84	0.83	0.83	0.81	0.86	0.71	1	
MAX	0.89	0.53	0.52	0.52	0.51	0.54	0.68	0.51	1

Panel C: Correlation matrix between IVOL and other factors

	IVOL	BETA	Size	Reversal	MOM	ILLI	ISKEW	Ratio	Lead_IVOL	MAX
IVOL	1									
BETA	0.09	1								
Size	-0.42	0.08	1							
Reversal	0.2	0.03	-0.01	1						
MOM	-0.07	0.03	0.12	-0.02	1					
ILLI	0.34	-0.03	-0.31	0	-0.07	1				
ISKEW	0.18	-0.01	-0.2	0.08	0.2	0.08	1			
Ratio	0.22	-0.02	-0.09	0.19	-0.02	0.02	0.16	1		
Lead_IVOL	0.67	0.08	-0.47	-0.01	-0.04	0.38	0.14	0.01	1	
MAX	0.89	0.12	-0.32	0.38	-0.06	0.25	0.21	0.37	0.51	1

Table 18. Average Monthly Correlation Matrices: IVOL Forecast Underestimation

This table presents average monthly correlation matrices for the subset of firm-months where forecasted IVOL falls below the realized IVOL in month $t+1$ (i.e., underestimation). The goal is to contrast the structure of misforecasted volatility across underprediction regimes. Panel A covers the interplay between various realized and event-based IVOL metrics. Panel B highlights the internal coherence among forecast models and their predictive accuracy relative to future volatility. Panel C examines how IVOL correlates with firm fundamentals and market microstructure characteristics during periods of underforecasting. All correlations are computed monthly and averaged across time.

Panel A: Correlation matrix between different IVOL metrics

	IVOL	IVOL_2	IVOL_3	IVOL_5	EAD_2	EAD_3	EAD_5	Maxadj	MAX Day	Lead_IVOL	MAX
IVOL	1										
IVOL_2	0.98	1									
IVOL_3	0.98	0.99	1								
IVOL_5	0.98	0.99	0.99	1							
EAD_2	0.78	0.51	0.49	0.44	1						
EAD_3	0.81	0.59	0.53	0.47	0.95	1					
EAD_5	0.87	0.75	0.72	0.52	0.85	0.89	1				
Maxadj	0.93	0.91	0.91	0.91	0.73	0.76	0.82	1			
EAD_max	0.91	0.88	0.87	0.88	0.68	0.71	0.77	0.69	1		
Lead_IVOL	0.61	0.61	0.61	0.61	0.45	0.48	0.53	0.62	0.48	1	
MAX	0.89	0.87	0.87	0.87	0.67	0.71	0.76	0.7	0.94	0.47	1

Panel B: Correlation matrix between IVOL and IVOL forecasts

	IVOL	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	Cross_F	Lead_IVOL	MAX
IVOL	1								
IVOL_F	0.68	1							
IVOL_2F	0.68	0.99	1						
IVOL_3F	0.68	0.98	0.99	1					
IVOL_5F	0.67	0.96	0.96	0.96	1				
Maxadj_F	0.68	0.96	0.96	0.96	0.94	1			
Cross_F	0.92	0.74	0.75	0.75	0.73	0.74	1		
Lead_IVOL	0.60	0.76	0.76	0.76	0.74	0.75	0.66	1	
MAX	0.89	0.53	0.53	0.53	0.52	0.53	0.71	0.47	1

Panel C: Correlation matrix between IVOL and other factors

	IVOL	BETA	Size	Reversal	MOM	ILLI	ISKEW	Ratio	Lead_IVOL	MAX
IVOL	1									
BETA	0.05	1								
Size	-0.46	0.08	1							
Reversal	0.14	0.01	-0.01	1						
MOM	-0.16	0.02	0.19	0.03	1					
ILLI	0.36	-0.03	-0.32	-0.01	-0.11	1				
ISKEW	0.21	0.02	-0.21	0.15	0.17	0.08	1			
Ratio	0.23	-0.01	-0.12	0.18	-0.03	0.02	0.21	1		
Lead_IVOL	0.6	0.01	-0.51	-0.05	-0.15	0.37	0.16	0.06	1	
MAX	0.89	0.09	-0.36	0.33	-0.13	0.27	0.28	0.38	0.47	1

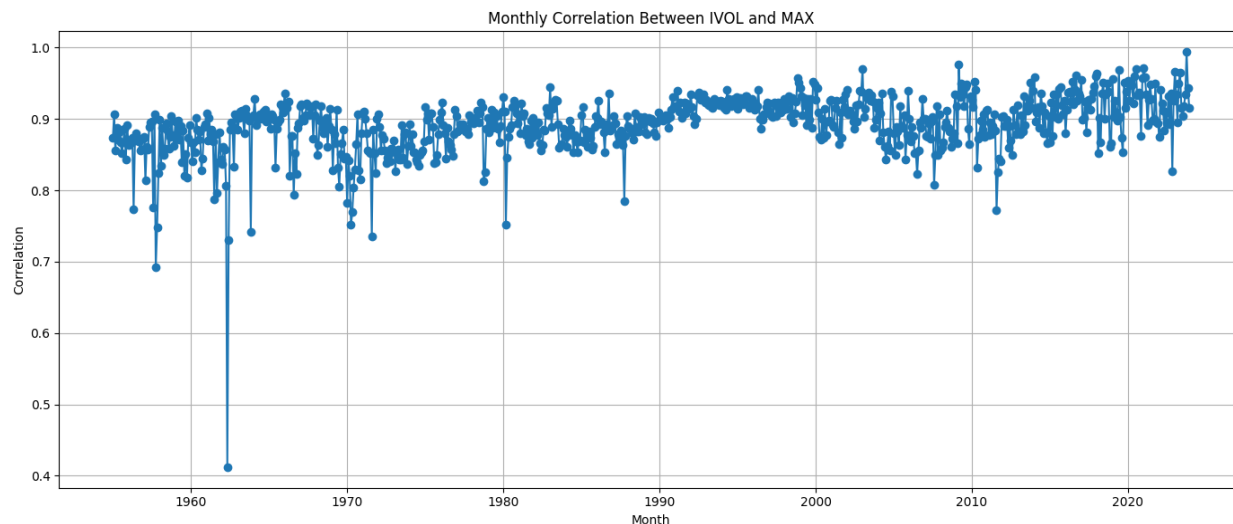


Figure 1. Monthly correlation between $MAX(t)$ and $IVOL(t)$, 1962–2023.

This figure plots the time series of monthly cross-sectional correlations between the maximum daily return in month t , (MAX_t), and the lead idiosyncratic volatility in month t , ($IVOL_t$). The sample covers the period from January 1955 to December 2023.

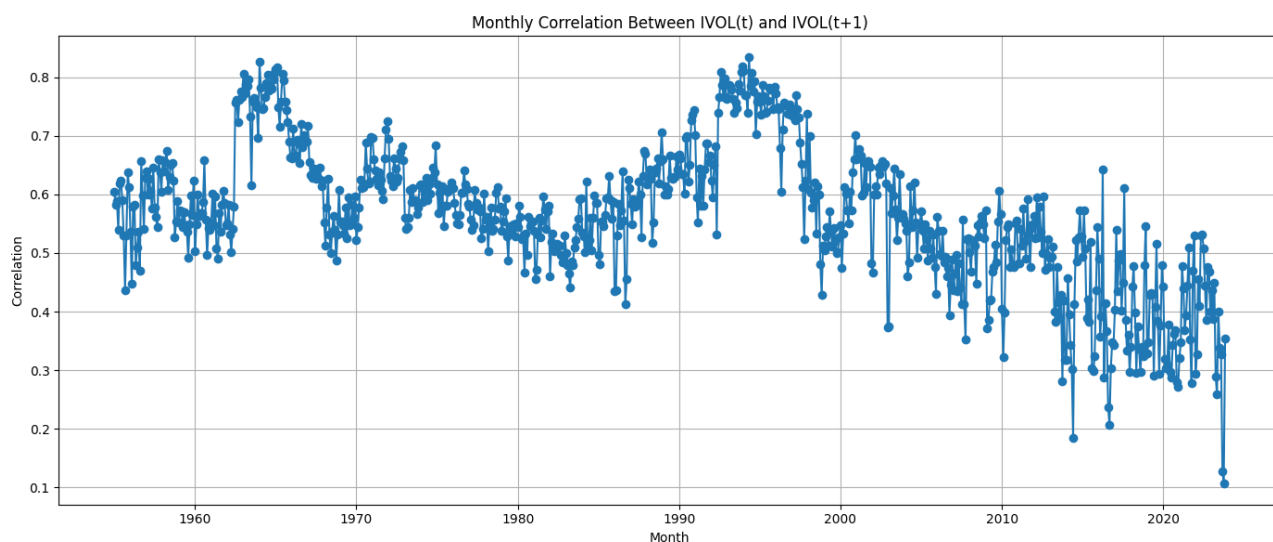


Figure 2. Monthly correlation between $IVOL(t)$ and $IVOL(t+1)$, 1962–2023.

This figure plots the time series of monthly cross-sectional correlations between the idiosyncratic volatility in month t , ($IVOL_t$), and the lead idiosyncratic volatility in month $t+1$, ($IVOL_{t+1}$). The sample covers the period from January 1955 to December 2023.

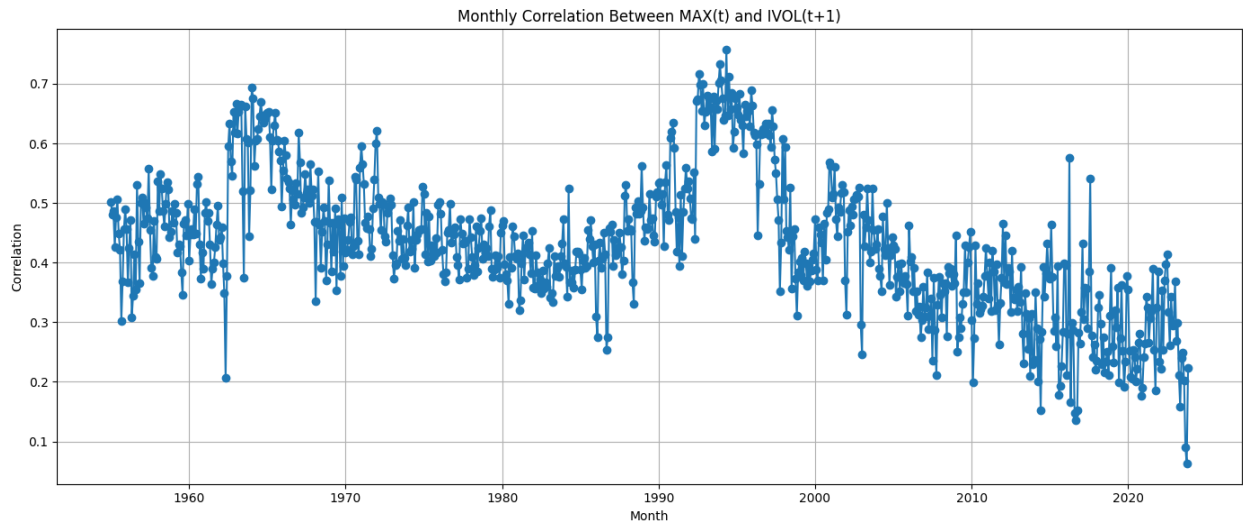


Figure 3. Monthly correlation between $MAX(t)$ and $IVOL(t+1)$, 1962–2023.

This figure plots the time series of monthly cross-sectional correlations between the maximum daily return in month t , (MAX_t), and the lead idiosyncratic volatility in month $t+1$, ($IVOL_{t+1}$). The sample covers the period from January 1955 to December 2023.

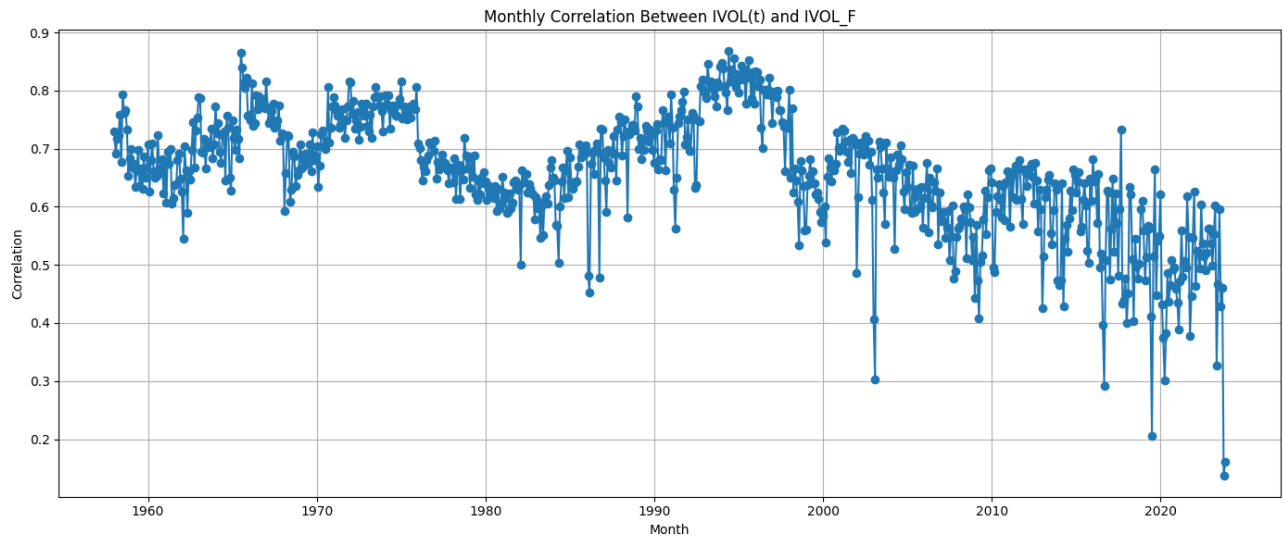


Figure 4. Monthly correlation between $IVOL(t)$ and ARMA (1,1) base Forecasted $IVOL$, 1962–2023.

This figure plots the time series of monthly cross-sectional correlations between the idiosyncratic volatility in month t , ($IVOL_t$), and the forecasted idiosyncratic volatility based on ARMA(1,1) in month t , ($IVOL_F$). The sample covers the period from January 1955 to December 2023.

Table 19. Temporal Variation in Monthly Correlations with Future Idiosyncratic Volatility

This table reports the average monthly correlations between various realized or forecasted idiosyncratic volatility (IVOL) measures and next-month realized IVOL (Lead_IVOL) over two subperiods: 1962–1992 and 1993–2023. The Δ Correlation column shows the change in mean correlation between the early and late periods. Values in parentheses are *z-statistics* from Fisher’s r-to-z transformation testing the null hypothesis that the two correlations are equal. Forecasted IVOL measures include ARMA(1,1) applied to realized IVOLs (IVOL_F), earnings-adjusted ARMA(1,1) forecasts (IVOL_2F, IVOL_3F, IVOL_5F), MAX-adjusted forecasts (Maxadj_F), and a cross-sectional model (IVOL_CF). MAX refers to the previous month’s maximum daily return.

Variable	Corr(1962–1992)	Corr(1993–2023)	Δ Correlation
IVOL	0.61	0.54	-0.08 (-8.78)
IVOL_F	0.68	0.61	-0.07 (-11.07)
IVOL_2F	0.68	0.61	-0.07 (-10.99)
IVOL_3F	0.68	0.61	-0.07 (-11.05)
IVOL_5F	0.67	0.58	-0.09 (-13.21)
Maxadj_F	0.69	0.64	-0.05 (-8.67)
IVOL_CF	0.65	0.60	-0.05 (-5.92)
MAX	0.47	0.40	-0.07 (-7.67)

Table 20. Forecast Errors by MAX Quintile

This table reports the forecasting errors of six IVOL models across quintiles of MAX, the stock's maximum daily return in the prior month, which proxies for episodic return-driven noise. In each month from July 1962 to December 2023, stocks are sorted into quintiles based on their lagged MAX values, and the root-mean-square error (RMSE) of out-of-sample IVOL forecasts is computed for each group. Panel A reports the median RMSE, while Panel B shows the mean RMSE.

Quintile	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Panel A: Median RMSE						
Low MAX	0.00292	0.00287	0.00287	0.00289	0.00256	0.00548
2	0.00348	0.00337	0.00336	0.00341	0.00317	0.00466
3	0.00466	0.00448	0.00447	0.00454	0.00427	0.00550
4	0.00629	0.00606	0.00604	0.00615	0.00581	0.00686
High MAX	0.00996	0.00974	0.00974	0.00990	0.00951	0.01037
Panel A: Mean RMSE						
Low MAX	0.01006	0.00997	0.00994	0.01005	0.00922	0.01292
2	0.00827	0.00823	0.00824	0.00837	0.00818	0.00977
3	0.01081	0.01076	0.01076	0.01097	0.01079	0.01207
4	0.01519	0.01518	0.01520	0.01552	0.01528	0.01612
High MAX	0.02922	0.02933	0.02929	0.03020	0.02881	0.02852

Table 21. Percentage of Over- and Underestimation Across MAX Quintiles

This table reports the average monthly percentages of stocks with overestimated (Panel A) and underestimated (Panel B) idiosyncratic volatility (IVOL) forecasts across quintiles sorted by the prior month's MAX return. For each month, stocks are sorted into five portfolios based on lagged MAX. Within each MAX quintile, we compute the percentage of firms whose forecasted IVOL exceeds the realized IVOL (overestimation) or falls below it (underestimation). The values shown are time-series averages of these monthly percentages. Columns represent different forecasting models: baseline forecast (IVOL_F), earnings-announcement adjusted forecasts with 2, 3, and 5 days (IVOL_2F, IVOL_3F, IVOL_5F), MAX-adjusted forecast (Maxadj_F), and cross-sectional model (IVOL_CF).

Quintile	IVOL_F	IVOL_2F	IVOL_3F	IVOL_5F	Maxadj_F	IVOL_CF
Panel A: Overestimate						
Low MAX	65.59	64.35	64.07	64.04	54.20	61.93
2	59.50	56.70	56.20	56.11	45.43	54.78
3	57.84	54.47	53.86	53.79	42.93	52.05
4	56.37	53.13	52.51	52.37	40.59	49.72
High MAX	52.59	50.04	49.51	49.23	36.86	50.41
Panel B: Underestimate						
Low MAX	34.41	35.65	35.93	35.96	45.80	38.07
2	40.50	43.30	43.80	43.89	54.57	45.22
3	42.16	45.53	46.14	46.21	57.07	47.95
4	43.63	46.87	47.49	47.63	59.41	50.28
High MAX	47.41	49.96	50.49	50.77	63.14	49.59

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